# Mathematical Statistics II

Lecture notes based on Kim, *Mathematical Statistics*, Ch. 6–11

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# Chapter 6. Estimation

## Setting

$X_1, \ldots, X_n$ is a random sample from $f(x;\theta)$, $\theta \in \Omega$. A statistic $\hat{\eta} = \hat{\eta}(X_1, \ldots, X_n)$ used to guess $\eta = \eta(\theta)$ is an **estimator**; its value at the data is an **estimate**. $P_\theta$, $E_\theta$, $\mathrm{Var}_\theta$ are computed under parameter $\theta$.

**Definition (Consistency)**

$\hat{\eta}_n$ is **consistent** for $\eta(\theta)$ if $\hat{\eta}_n \xrightarrow{p} \eta(\theta)$ under $P_\theta$ for every $\theta \in \Omega$, i.e. $\lim_n P_\theta(|\hat{\eta}_n - \eta(\theta)| \geq \epsilon) = 0$ for all $\epsilon > 0$ and all $\theta$.

## 6.1 The method of moments

The population moment $m_r = EX_1^r$ is estimated by $\hat{m}_r = \frac{1}{n}\sum X_i^r$. If $\eta = g(m_1, \ldots, m_k)$, the **method-of-moments estimator** (MME) is

$$\hat{\eta}^{\mathrm{MME}} = g(\hat{m}_1, \ldots, \hat{m}_k).$$

Multivariate populations use joint moments $m_{r,s} = E(X_1^r Y_1^s)$ and $\hat{m}_{r,s} = \frac{1}{n}\sum X_i^r Y_i^s$.

**Example 6.1**

(a) $\sigma^2 = m_2 - m_1^2$, so

$$\hat{\sigma}^{2\,\mathrm{MME}} = \hat{m}_2 - \hat{m}_1^2 = \frac{1}{n}\sum_{i=1}^n (X_i - \bar{X})^2, \qquad \hat{\sigma}^{\mathrm{MME}} = \sqrt{\frac{1}{n}\sum (X_i - \bar{X})^2}.$$

(b) The MME is not unique: for Poisson($\lambda$), $\lambda = m_1 = m_2 - m_1^2$ gives $\bar{X}$ and $\frac{1}{n}\sum(X_i - \bar{X})^2$. Use the lowest moments possible.

(c) For a bivariate population, $\rho = \dfrac{m_{1,1} - m_{1,0}m_{0,1}}{\sqrt{m_{2,0} - m_{1,0}^2}\sqrt{m_{0,2} - m_{0,1}^2}}$, so

$$\hat{\rho}^{\mathrm{MME}} = \frac{\sum (X_i - \bar{X})(Y_i - \bar{Y})}{\sqrt{\sum (X_i - \bar{X})^2}\sqrt{\sum (Y_i - \bar{Y})^2}},$$

the sample correlation coefficient.

(d) Gamma($\alpha, \beta$): $m_1 = \alpha\beta$, $m_2 - m_1^2 = \alpha\beta^2$ give $\hat{\beta} = \hat{\sigma}^2/\bar{X}$, $\hat{\alpha} = \bar{X}^2/\hat{\sigma}^2$, $\hat{\sigma}^2 = \frac{1}{n}\sum(X_i - \bar{X})^2$.

**Theorem 6.1 (Consistency of MMEs)**

If $E|X_1|^k < \infty$ and $g$ is continuous at $(m_1, \ldots, m_k)$, then $\hat{\eta}_n^{\mathrm{MME}} = g(\hat{m}_1, \ldots, \hat{m}_k) \xrightarrow{p} g(m_1, \ldots, m_k) = \eta$.

**Proof.** By the multivariate LLN $(\hat{m}_1, \ldots, \hat{m}_k)' \xrightarrow{p} (m_1, \ldots, m_k)'$ (Example 5.5 in Volume I), and continuous maps preserve convergence in probability (Theorem 5.7 in Volume I). ■

So $\hat{\sigma}^{2\,\mathrm{MME}}$, $S^2 = \frac{n}{n-1}\hat{\sigma}^{2\,\mathrm{MME}}$ and the sample correlation are consistent. MMEs are also asymptotically normal:

**Theorem 6.2 (Limiting distribution of MMEs)**

If $EX_1^{2k} < \infty$ and $g$ has continuous first partial derivatives at $\mathbf{m} = (m_1, \dots, m_k)'$, then

$$\sqrt{n}(\hat{\eta}_n^{\text{MME}} - \eta) \xrightarrow{d} N(0,\ \dot{g}(\mathbf{m})'\Sigma\,\dot{g}(\mathbf{m})), \qquad \Sigma = (m_{r+s} - m_r m_s)_{r,s=1}^{k}.$$

**Proof.** $(X_i, X_i^2, \dots, X_i^k)'$ are i.i.d. with mean $\mathbf{m}$ and covariance $\Sigma$ ($\text{Cov}(X_1^r, X_1^s) = m_{r+s} - m_r m_s$); apply the multivariate CLT and the delta method (Theorem 5.12 in Volume I). ■

**Example 6.2**

$g(m_1, m_2) = m_2 - m_1^2$, $\dot{g} = (-2m_1, 1)'$, and $\dot{g}'\Sigma\dot{g} = \mu_4 - \sigma^4$, $\mu_4 = E(X_1 - \mu)^4$ (cf. Example 5.10 in Volume I).

## 6.2 Maximum likelihood

The sample falls in a small box around $\mathbf{x}$ with probability about $\prod_i f(x_i;\theta)\,|\Delta\mathbf{x}|$, so comparing $\prod f(x_i;\theta)$ across $\theta$ ranks parameter values by plausibility: regard the joint pdf as a function of $\theta$.

**Definition (Likelihood and MLE)**

(a) The **likelihood function** is $L(\theta) = L(\theta;\mathbf{x}) = \prod_{i=1}^n f(x_i;\theta)$, $\theta \in \Omega$, and $l(\theta) = \log L(\theta) = \sum \log f(x_i;\theta)$ is the **log-likelihood**.

(b) A value $\hat{\theta}(\mathbf{x}) \in \Omega$ with $L(\hat{\theta}(\mathbf{x});\mathbf{x}) = \max_{\theta\in\Omega} L(\theta;\mathbf{x})$ is a **maximum likelihood estimate**; the statistic $\hat{\theta}^{\text{MLE}} = \hat{\theta}(\mathbf{X})$ is the **maximum likelihood estimator** (MLE).

$\dot{l}(\theta) = 0$ is the **likelihood equation**; the MLE is usually among its roots.

**Example 6.3 (Poisson)**

Poisson($\lambda$), $\lambda \geq 0$ ($\lambda = 0$: point mass at 0):

$$l(\lambda) = -n\lambda + n\bar{x}\log\lambda - \log(x_1!\cdots x_n!), \qquad \dot{l}(\lambda) = -n + \frac{n\bar{x}}{\lambda}, \qquad \ddot{l}(\lambda) = -\frac{n\bar{x}}{\lambda^2}.$$

If $\bar{x} > 0$, $l$ increases up to $\bar{x}$ and then decreases; if $\bar{x} = 0$, $l = -n\lambda$ decreases. By continuity of $L$ on $[0,\infty)$, $\hat{\lambda}^{\text{MLE}} = \bar{X}$ in both cases.

**Theorem 6.3**

Let $l$ be twice differentiable with continuous second derivative on an open interval $\Omega_o$. If $\ddot{l}(\theta) < 0$ for all $\theta \in \Omega_o$ and $\dot{l}(\hat{\theta}) = 0$ for some $\hat{\theta} \in \Omega_o$, then $\hat{\theta}$ is the unique maximizer of $l$ on $\Omega_o$.

**Proof.** By Taylor, with $\theta^*$ between $\theta$ and $\hat{\theta}$,

$$l(\theta) = l(\hat{\theta}) + \dot{l}(\hat{\theta})(\theta - \hat{\theta}) + \tfrac{1}{2}\ddot{l}(\theta^*)(\theta - \hat{\theta})^2 = l(\hat{\theta}) + \tfrac{1}{2}\ddot{l}(\theta^*)(\theta - \hat{\theta})^2 < l(\hat{\theta})$$

for $\theta \neq \hat{\theta}$. ■

**Example 6.4 (Bernoulli)**

Bernoulli$(p)$, $0 \le p \le 1$, $\hat{p} = \bar{x}$:

$$l(p) = n\hat{p}\log p + n(1-\hat{p})\log(1-p), \qquad \dot{l}(p) = \frac{n\hat{p}}{p} - \frac{n(1-\hat{p})}{1-p}, \qquad \ddot{l}(p) = -\frac{n\hat{p}}{p^2} - \frac{n(1-\hat{p})}{(1-p)^2}.$$

If $0 < \hat{p} < 1$, $\ddot{l} < 0$ and $\dot{l}(\hat{p}) = 0$ (Theorem 6.3); if $\hat{p} = 0$, $L = (1-p)^n$ peaks at 0; if $\hat{p} = 1$, $L = p^n$ at 1. So $\hat{p}^{\mathrm{MLE}} = \bar{X}$.

$\ddot{l} < 0$ means $l$ is **strictly concave**. When the likelihood equation has no explicit solution:

**Theorem 6.4**

Let $l$ be twice continuously differentiable on an open interval $\Omega_o$, with $\ddot{l}(\theta) < 0$ on $\Omega_o$ and $l(\theta) \to -\infty$ as $\theta$ tends to either end of $\Omega_o$. Then the likelihood equation $\dot{l}(\theta) = 0$ has exactly one root $\hat{\theta}$ in $\Omega_o$, and $\hat{\theta}$ is the unique maximizer of $l$.

**Proof.** Take $\Omega_o = \mathbb{R}$ (others similar). As $l \to -\infty$ at both ends, $l$ attains its maximum; at two points $\hat{\theta}_1 < \hat{\theta}_2$ strict concavity would give

$$l\left(\tfrac{1}{2}\hat{\theta}_1 + \tfrac{1}{2}\hat{\theta}_2\right) > \tfrac{1}{2}l(\hat{\theta}_1) + \tfrac{1}{2}l(\hat{\theta}_2) = \max l,$$

a contradiction. At the maximum $\dot{l}(\hat{\theta}) = 0$, and $\dot{l}$ is strictly decreasing. ■

Without a closed form, use **Newton–Raphson**: from $0 = \dot{l}(\hat{\theta}) \approx \dot{l}(\theta^{(r)}) + \ddot{l}(\theta^{(r)})(\hat{\theta} - \theta^{(r)})$,

$$\hat{\theta}^{(r+1)} = \hat{\theta}^{(r)} + \left[-\ddot{l}(\hat{\theta}^{(r)})\right]^{-1}\dot{l}(\hat{\theta}^{(r)}), \qquad r = 0, 1, \ldots,$$

started e.g. at the MME.

**Example 6.5 (Logistic location)**

$f(x;\theta) = e^{-(x-\theta)}/(1+e^{-(x-\theta)})^2$, $\theta \in \mathbb{R}$:

$$l(\theta) = -n\bar{x} + n\theta - 2\sum_{i=1}^{n}\log\left(1 + e^{-(x_i-\theta)}\right), \qquad \dot{l}(\theta) = n - 2\sum_i \frac{e^{-(x_i-\theta)}}{1+e^{-(x_i-\theta)}},$$

$$\ddot{l}(\theta) = -2\sum_i \frac{e^{-(x_i-\theta)}}{(1+e^{-(x_i-\theta)})^2} < 0.$$

As $\theta \to -\infty$, $l \approx n\theta - n\bar{x} \to -\infty$; as $\theta \to \infty$, $l \approx -n\theta + n\bar{x} \to -\infty$. By Theorem 6.4 the MLE is the unique root; start Newton–Raphson at $\bar{x}$ (the logistic mean is $\theta$).

**Theorem 6.5 (Invariance)**

If $\hat{\theta}^{\mathrm{MLE}}$ exists and $\eta = g(\theta)$ is a one-to-one function on $\Omega$, then $\hat{\eta}^{\mathrm{MLE}} = g(\hat{\theta}^{\mathrm{MLE}})$.

**Proof.** The density with parameter $\eta$ is $f(\cdot; g^{-1}(\eta))$, so $L^*(\eta) = L(g^{-1}(\eta))$ and

$$\max_{\eta \in g(\Omega)} L^*(\eta) = \max_{\theta \in \Omega} L(\theta) = L(\hat{\theta}^{\mathrm{MLE}}) = L^*(g(\hat{\theta}^{\mathrm{MLE}})).$$

■

**Example 6.6 (Exponential)**

$f(x;\theta)=\theta^{-1}e^{-x/\theta}$: $l(\theta)=-n\log\theta-n\bar{x}/\theta$ has $\ddot{l}(\theta)=n/\theta^2-2n\bar{x}/\theta^3$, not always negative. With $\lambda=1/\theta$, $l=n\log\lambda-n\bar{x}\lambda$ is strictly concave and $\to-\infty$ at $0,\infty$ ($\bar{x}>0$): $\hat{\lambda}=1/\bar{X}$, so $\hat{\theta}^{\mathrm{MLE}}=\bar{X}$ by invariance.

A good parametrization can make $l$ concave; exponential families have a canonical one.

**Theorem 6.6 (MLE in a one-parameter exponential family)**

Let

$$f(x;\eta)=\exp\{\eta T(x)-A(\eta)+S(x)\},\qquad x\in\mathcal{X},\ \eta\in N,$$

where (i) the support $\mathcal{X}$ does not depend on $\eta$;
(ii) $N$ is an open interval;
(iii) $\mathrm{Var}_\eta\, T(X_1)>0$. If the likelihood equation

$$\dot{A}(\eta)=\frac{1}{n}\sum_{i=1}^{n}T(x_i),\qquad \text{i.e.}\quad E_\eta T(X_1)=\frac{1}{n}\sum_{i=1}^{n}T(x_i),$$

has a root $\hat{\eta}$ in $N$, then $\hat{\eta}$ is the MLE of $\eta$.

**Proof.** The cgf of $T(X_1)$: for small $|s|$ ($s+\eta\in N$ by (ii)), in the continuous case

$$E_\eta e^{sT(X_1)}=\int e^{(s+\eta)T(x)-A(\eta)+S(x)}dx=e^{A(s+\eta)-A(\eta)}\int f(x;s+\eta)\,dx=e^{A(s+\eta)-A(\eta)},$$

using (i). So $E_\eta T(X_1)=\dot{A}(\eta)$ and $\mathrm{Var}_\eta\, T(X_1)=\ddot{A}(\eta)>0$ (iii). Then $l(\eta)=\eta\sum T(x_i)-nA(\eta)+\sum S(x_i)$ has $\ddot{l}=-n\ddot{A}<0$, and a root of $\dot{l}=\sum T(x_i)-n\dot{A}(\eta)$ is the MLE (Theorem 6.3). ■

In another parameter, $\eta=g(\theta)$ one-to-one, invariance gives: *match the mean of $T$ to its sample average*, $E_\theta T(X_1)=\frac{1}{n}\sum T(x_i)$.

**Example 6.7**

(a) Bernoulli($p$): $f=\exp\{x\log\frac{p}{1-p}+\log(1-p)\}$, $T(x)=x$; $\hat{p}=\bar{x}$ for $0<\bar{x}<1$.

(b) Poisson($\lambda$): $T(x)=x$, $\eta=\log\lambda$; $\hat{\lambda}=\bar{x}$ when $\bar{x}>0$.

(c) Geometric $(1-p)^{x-1}p$, $x\geq 1$: $T(x)=x$, $\eta=\log(1-p)$; $E_pX=1/p=\bar{x}$ gives $\hat{p}=1/\bar{x}$ ($\bar{x}>1$).

(d) Exponential($\theta$): $T(x)=x$, $\eta=-1/\theta$; $\hat{\theta}=\bar{x}$.

(e) Pareto $\theta x^{-\theta-1}$, $x>1$: $T(x)=-\log x$; $\theta\log X\sim\mathrm{Exp}(1)$ (as $X^{-\theta}\sim U(0,1)$), so $E_\theta(-\log X)=-1/\theta$ and $\hat{\theta}=1/(\frac{1}{n}\sum\log x_i)$.

Without differentiability, study $L$ directly; the MLE need not be unique.

**Example 6.8 (Double exponential)**

$f(x;\theta)=\frac{1}{2}e^{-|x-\theta|}$, $n=2m+1$: with $x_{(0)}=-\infty$, $x_{(n+1)}=\infty$, for $x_{(r)}\leq\theta<x_{(r+1)}$

$$l(\theta)=(n-2r)\theta+\sum_{i\leq r}x_{(i)}-\sum_{i>r}x_{(i)}-n\log 2.$$

The slope $n-2r$ is positive for $r\leq m$, negative for $r\geq m+1$: the MLE is the median $X_{(m+1)}$.

**Example 6.9 (Uniform)**

(a) $U(0,\theta)$: $L(\theta)=\theta^{-n}\mathbf{1}\{\theta\ge x_{(n)}\}$, so $\hat\theta^{\mathrm{MLE}}=X_{(n)}$ (no likelihood equation).
(b) $U(\theta-1,\theta+1)$: $L(\theta)=2^{-n}\mathbf{1}\{x_{(n)}-1\le\theta\le x_{(1)}+1\}$, so every statistic in $[X_{(n)}-1, X_{(1)}+1]$, e.g. $\alpha(X_{(n)}-1)+(1-\alpha)(X_{(1)}+1)$, is an MLE; other criteria must choose.

## 6.3 MLE for vector parameters

For $\theta\in\Omega\subseteq\mathbb{R}^k$ the definitions are the same; components of $\hat\theta^{\mathrm{MLE}}$ are MLEs of the components. Maximize one variable at a time, or solve $\partial l/\partial\theta_j=0$.

**Example 6.10 (Double exponential with scale)**

$f(x;\mu,\sigma)=\frac{1}{2\sigma}e^{-|x-\mu|/\sigma}$, $n=2m+1$:

$$l(\mu,\sigma)=-\frac{1}{\sigma}\sum|x_i-\mu|-n\log\sigma-n\log 2.$$

For each $\sigma$ the median $\hat\mu=x_{(m+1)}$ maximizes in $\mu$ (Example 6.8), and

$$\frac{\partial}{\partial\sigma}l(\hat\mu,\sigma)=\frac{\sum|x_i-\hat\mu|}{\sigma^2}-\frac{n}{\sigma}>0\iff\sigma<\frac{1}{n}\sum|x_i-\hat\mu|,$$

so $\hat\sigma=\frac{1}{n}\sum|x_i-\hat\mu|$; the MLE is $(X_{(m+1)},\frac{1}{n}\sum|X_i-X_{(m+1)}|)$.

**Example 6.11 (Normal)**

$N(\mu,\sigma^2)$:

$$l(\mu,\sigma^2)=-\frac{1}{2\sigma^2}\sum(x_i-\mu)^2-\frac{n}{2}\log\sigma^2-\frac{n}{2}\log 2\pi.$$

$\sum(x_i-\mu)^2=\sum(x_i-\bar x)^2+n(\bar x-\mu)^2$, so $\mu=\bar x$ for every $\sigma^2$; then $\partial_{\sigma^2}l(\bar x,\sigma^2)=\frac{1}{2\sigma^4}\sum(x_i-\bar x)^2-\frac{n}{2\sigma^2}>0$ iff $\sigma^2<\frac{1}{n}\sum(x_i-\bar x)^2$. Hence $\hat\mu=\bar X$, $\hat\sigma^2=\frac{1}{n}\sum(X_i-\bar X)^2$.

**Theorem 6.7 (Invariance for functions)**

If $\hat\theta^{\mathrm{MLE}}$ exists and $\eta_1=g_1(\theta)$ can be completed to a one-to-one map $\eta=(g_1(\theta),g_2(\theta))$, then $\hat\eta_1^{\mathrm{MLE}}=g_1(\hat\theta^{\mathrm{MLE}})$.

**Proof.** By Theorem 6.5 (verbatim for vectors), $\hat\eta^{\mathrm{MLE}}=(g_1(\hat\theta^{\mathrm{MLE}}),g_2(\hat\theta^{\mathrm{MLE}}))$; take the first component. ■

**Example 6.12**

(a) $N(\mu,\sigma^2)$: $(|\mu|,\operatorname{sgn}\mu,\sigma^2)$ is one-to-one, so $\widehat{|\mu|}^{\mathrm{MLE}}=|\bar X|$.
(b) Bernoulli: $(p(1-p),\operatorname{sgn}(p-\frac12))$ is one-to-one, so $\hat p(1-\hat p)$ is the MLE of $p(1-p)$.

The concavity results extend (same proofs):

**Theorem 6.8**

Let $\Omega_o = (a_1, b_1) \times \cdots \times (a_k, b_k)$ and let $l$ have continuous second partial derivatives, with Hessian $\ddot{l}(\theta) = (\partial^2 l / \partial\theta_i \partial\theta_j)$.

(a) If $\mathbf{c}'\ddot{l}(\theta)\mathbf{c} < 0$ for all $\mathbf{c} \neq \mathbf{0}$ and $\theta \in \Omega_o$, and $\dot{l}(\hat{\theta}) = \mathbf{0}$ for some $\hat{\theta} \in \Omega_o$, then $\hat{\theta}$ is the unique maximizer of $l$.

(b) If, moreover, $l(\theta) \to -\infty$ as $\theta$ tends to the boundary of $\Omega_o$, then $\dot{l}(\theta) = \mathbf{0}$ has exactly one root in $\Omega_o$, and it is the MLE.

**Theorem 6.9 (Multiparameter exponential family)**

Let $f(x;\eta) = \exp\{\sum_{j=1}^k \eta_j T_j(x) - A(\eta) + S(x)\}$, $x \in \mathcal{X}$, $\eta \in N$, where (i) $\mathcal{X}$ does not depend on $\eta$;
(ii) $N$ contains an open box in $\mathbb{R}^k$;
(iii) $\mathrm{Var}_\eta(\mathbf{c}'\mathbf{T}(X_1)) > 0$ for every $\mathbf{c} \neq \mathbf{0}$. If the likelihood equations

$$E_\eta T_j(X_1) = \frac{1}{n}\sum_{i=1}^n T_j(x_i), \qquad j = 1, \ldots, k,$$

have a root $\hat{\eta}$ in $N$, then $\hat{\eta}$ is the MLE.

**Proof.** As in Theorem 6.6, $E_\eta \mathbf{T} = \dot{A}(\eta)$, $\mathrm{Var}_\eta \mathbf{T} = \ddot{A}(\eta)$, so $\mathbf{c}'\ddot{l}(\eta)\mathbf{c} = -n\,\mathrm{Var}_\eta(\mathbf{c}'\mathbf{T}) < 0$; apply Theorem 6.8(a). ■

**Example 6.13**

(a) *Normal*: $\eta_1 = \mu/\sigma^2$, $\eta_2 = -1/(2\sigma^2)$, $T = (x, x^2)$; matching gives $\hat{\mu} = \bar{x}$, $\hat{\sigma}^2 + \hat{\mu}^2 = \frac{1}{n}\sum x_i^2$, as in Example 6.11.
(b) *Bivariate normal*: $T = (x, y, xy, x^2, y^2)$; matching gives $\hat{\mu}_1 = \bar{x}$, $\hat{\mu}_2 = \bar{y}$, $\hat{\sigma}_j^2 = \frac{1}{n}\sum(\cdot - \text{mean})^2$, and $\hat{\rho}^{\text{MLE}}$ the sample correlation.
(c) *Multinomial* $\text{Multi}(n, \mathbf{p})$, one observation: with $\eta_j = \log(p_j/p_k)$, $j < k$, the pmf is $\exp\{\sum_{j<k}\eta_j x_j - nA(\eta) + S(x)\}$, $A(\eta) = \log(1 + e^{\eta_1} + \cdots + e^{\eta_{k-1}})$; $EX_j = np_j = x_j$ gives $\hat{p}_j = x_j/n$ (all $0 < x_j < n$).

## 6.4 Asymptotic properties of the MLE

**Motivation.** Under regularity the MLE is consistent and asymptotically normal with the Cramér–Rao variance.

**Example 6.14**

When the MLE is an MME, Chapter 5 applies:

- Bernoulli: $\hat{p} \xrightarrow{p} p$ and $\sqrt{n}(\hat{p} - p) \xrightarrow{d} N(0, p(1-p))$.
- Poisson: $\sqrt{n}(\hat{\lambda} - \lambda) \xrightarrow{d} N(0, \lambda)$. Exponential: $\sqrt{n}(\hat{\theta} - \theta) \xrightarrow{d} N(0, \theta^2)$.
- Normal: $\sqrt{n}(\hat{\mu} - \mu) \sim N(0, \sigma^2)$ exactly, $n\hat{\sigma}^2/\sigma^2 \sim \chi^2(n-1)$, and $\sqrt{n}(\hat{\sigma}^2 - \sigma^2) \xrightarrow{d} N(0, 2\sigma^4)$.
- Bivariate normal: $\sqrt{n}(\hat{\rho} - \rho) \xrightarrow{d} N(0, (1-\rho^2)^2)$.

In general we argue directly. Let $\bar{l}_n(\theta) = \frac{1}{n}\sum_i \log f(X_i;\theta)$ and $\theta^0$ the true value. By the LLN $\bar{l}_n(\theta) \xrightarrow{p} E_{\theta^0} \log f(X_1;\theta)$, so $\hat{\theta}_n = \arg\max \bar{l}_n$ should approach $\arg\max_\theta E_{\theta^0} \log f(X_1;\theta)$, which is $\theta^0$:

**Theorem 6.10 (Kullback–Leibler divergence)**

Assume (R0) *identifiability*: $f(\cdot;\theta) = f(\cdot;\theta^0)$ implies $\theta = \theta^0$; and (R1) the support $\{x : f(x;\theta) > 0\} = \mathcal{X}$ does not depend on $\theta$. Let

$$KL(\theta,\theta^0) = -E_{\theta^0} \log \frac{f(X;\theta)}{f(X;\theta^0)}.$$

Then (i) $KL(\theta,\theta^0) \geq 0$ for all $\theta$, and (ii) $KL(\theta,\theta^0) = 0$ iff $\theta = \theta^0$. Equivalently, $\theta^0$ is the unique maximizer of $E_{\theta^0} \log f(X;\theta)$.

**Proof.** $-\log t \geq 1 - t$ $(t > 0)$, with equality iff $t = 1$. With $t = f(X;\theta)/f(X;\theta^0)$,

$$KL(\theta,\theta^0) \geq E_{\theta^0}\Big[1 - \frac{f(X;\theta)}{f(X;\theta^0)}\Big] = 1 - \int_{\mathcal{X}} f(x;\theta)\,dx = 0,$$

by (R1); equality iff $f(\cdot;\theta) = f(\cdot;\theta^0)$ a.e., i.e. $\theta = \theta^0$ by (R0). ∎

A proof needs e.g. uniform convergence of $\bar{l}_n$ on compacts and continuity of $E_{\theta^0} \log f(X_1;\theta)$; these hold under strict concavity with $l_n \to -\infty$ at the boundary:

**Theorem 6.11 (Consistency of the MLE)**

Let $\Omega$ be an open box in $\mathbb{R}^k$ and assume (R0), (R1). Suppose $l_n$ has continuous second partial derivatives with $\mathbf{c}'\ddot{l}_n(\theta)\mathbf{c} < 0$ for $\mathbf{c} \neq \mathbf{0}$ and $l_n(\theta) \to -\infty$ at the boundary of $\Omega$, and that $E_{\theta^0} \log f(X_1;\theta)$ exists and is continuous in $\theta$. Then the MLE $\hat{\theta}_n$ is the unique root of $\dot{l}_n(\theta) = \mathbf{0}$, and $\hat{\theta}_n \xrightarrow{p} \theta$ under $P_\theta$ for every $\theta \in \Omega$.

(Proof omitted.) E.g. the logistic MLE (Example 6.5) and the gamma MLE (root of $-\Psi(\alpha) - \log\beta + \overline{\log x} = 0$, $-\alpha/\beta + \bar{x}/\beta^2 = 0$, $\Psi = (\log\Gamma)'$) are consistent.

**Heuristics for the limit law.** Expanding the likelihood equation,

$$0 = \dot{\bar{l}}_n(\hat{\theta}_n) \approx \dot{\bar{l}}_n(\theta) + \ddot{\bar{l}}_n(\theta)(\hat{\theta}_n - \theta), \qquad \text{so} \qquad \sqrt{n}(\hat{\theta}_n - \theta) \approx \{-\ddot{\bar{l}}_n(\theta)\}^{-1}\sqrt{n}\,\dot{\bar{l}}_n(\theta).$$

and $-\ddot{\bar{l}}_n(\theta) \xrightarrow{p} I(\theta) := E_\theta[-\partial_\theta^2 \log f(X_1;\theta)]$ (LLN), while by the CLT

$$\sqrt{n}\,\dot{\bar{l}}_n(\theta) \xrightarrow{d} N\big(0, \mathrm{Var}_\theta\, \partial_\theta \log f(X_1;\theta)\big),$$

provided $E_\theta \partial_\theta \log f(X_1;\theta) = 0$.

**Regularity conditions.** In addition to (R0) and (R1):

(R2) $\Omega$ is an open subset of $\mathbb{R}^k$.

(R3) For every $x \in \mathcal{X}$, $\log f(x;\theta)$ has continuous first and second partial derivatives in $\theta$.

(R4) For $u(\mathbf{X})$ with $E_\theta u(\mathbf{X})$ finite, $\partial_\theta^r E_\theta u(\mathbf{X}) = \int u(\mathbf{x})\partial_\theta^r \prod f(x_i;\theta)\,d\mathbf{x}$, $r = 1, 2$.

(R5) The **Fisher information** $I(\theta) = \mathrm{Var}_\theta\,(\partial_\theta \log f(X_1;\theta))$ is finite and nonsingular.

$\dot{l}_n(\theta)$ is the **score function**.

**Theorem 6.12 (Properties of the score)**

Under (R0)–(R5),

(a) $E_\theta\{\partial_\theta \log f(X_1;\theta)\} = 0$;

(b) $I(\theta) = \mathrm{Var}_\theta\{\partial_\theta \log f(X_1;\theta)\} = E_\theta\{-\partial_\theta^2 \log f(X_1;\theta)\}$.

**Proof.** One-dimensional continuous $\theta$: $\int_{\mathcal{X}} \exp\{\log f(x;\theta)\}\,dx = 1$ (R1); differentiating under the integral (R3, R4),

$$\int \{\partial_\theta \log f(x;\theta)\} f(x;\theta)\,dx = 0, \qquad \text{i.e.} \quad E_\theta \partial_\theta \log f(X_1;\theta) = 0.$$

and again,

$$\int \left\{\partial_\theta^2 \log f(x;\theta) + \left[\partial_\theta \log f(x;\theta)\right]^2\right\} f(x;\theta)\,dx = 0,$$

so $E_\theta[(\partial_\theta \log f)^2] = -E_\theta[\partial_\theta^2 \log f]$, the variance by (a). ∎

**Theorem 6.13 (Asymptotic normality of the MLE)**

Assume (R0)–(R5) and also:

(R6) the MLE $\hat\theta_n$ is the unique root of $\dot l_n(\theta) = 0$ and is consistent;

(R7) the third derivatives $\partial_\theta^3 \log f(x;\theta)$ exist and are continuous, and there is $M(x)$ with $\max_{\theta\in\Omega} \|\partial_\theta^3 \log f(x;\theta)\| \le M(x)$ and $E_\theta M(X_1) < \infty$.

Then $\sqrt{n}(\hat\theta_n - \theta) \xrightarrow{d} N(0, I(\theta)^{-1})$.

**Proof.** One-dimensional; $\dot{\bar l}_n = \dot l_n/n$, $\ddot{\bar l}_n = \ddot l_n/n$, $\bar l_n^{(3)} = l_n^{(3)}/n$.

*Step 1 (expansion).* By Taylor, with $\theta_n^*$ between $\theta$ and $\hat\theta_n$,

$$0 = \dot{\bar l}_n(\hat\theta_n) = \dot{\bar l}_n(\theta) + \ddot{\bar l}_n(\theta)(\hat\theta_n - \theta) + \tfrac{1}{2}\bar l_n^{(3)}(\theta_n^*)(\hat\theta_n - \theta)^2.$$

*Step 2 (the remainder is negligible).* Let $R_n = \frac{1}{2}\bar l_n^{(3)}(\theta_n^*)(\hat\theta_n - \theta)$. For $\epsilon, K > 0$,

$$P_\theta(|R_n| \ge \epsilon) \le P_\theta(|\bar l_n^{(3)}(\theta_n^*)| > K) + P_\theta(|\hat\theta_n - \theta| \ge 2\epsilon/K).$$

By (R7) and Markov, $P_\theta(|\bar l_n^{(3)}(\theta_n^*)| > K) \le E_\theta M(X_1)/K$; by (R6) the second term $\to 0$. So $\limsup_n P_\theta(|R_n| \ge \epsilon) \le E_\theta M(X_1)/K$ for all $K$: $R_n \xrightarrow{p} 0$.

*Step 3 (linearization).* $0 = \dot{\bar l}_n(\theta) + \{\ddot{\bar l}_n(\theta) + R_n\}(\hat\theta_n - \theta)$ and $-\ddot{\bar l}_n(\theta) \xrightarrow{p} I(\theta)$ (LLN, Theorem 6.12(b)), so

$$\sqrt{n}(\hat\theta_n - \theta) = \big(I(\theta) + r_n\big)^{-1} \sqrt{n}\,\dot{\bar l}_n(\theta), \qquad r_n \xrightarrow{p} 0.$$

*Step 4 (limit).* $\sqrt{n}\,\dot{\bar l}_n(\theta) \xrightarrow{d} N(0, I(\theta))$ (CLT, Theorem 6.12); by Slutsky, $\sqrt{n}(\hat\theta_n - \theta) \xrightarrow{d} I(\theta)^{-1} Z \sim N(0, I(\theta)^{-1})$. ∎

(R7) is stronger than necessary. Step 3 also gives

$$\sqrt{n}(\hat\theta_n - \theta) = I(\theta)^{-1} \sqrt{n}\,\dot{\bar l}_n(\theta) + o_p(1):$$

the MLE error is driven by the score.

One Newton–Raphson step from a good start is asymptotically as good:

**Theorem 6.14 (One-step estimator)**

Assume (R0)–(R7), and let $\hat\theta_n^{(0)}$ be an estimator with $\sqrt{n}(\hat\theta_n^{(0)}-\theta)\xrightarrow{d} Z$ for some random variable $Z$ (R8). Then

$$\hat\theta_n^{(1)}=\hat\theta_n^{(0)}+\big[-\ddot l_n(\hat\theta_n^{(0)})\big]^{-1}\dot l_n(\hat\theta_n^{(0)})$$

satisfies $\sqrt{n}(\hat\theta_n^{(1)}-\theta)\xrightarrow{d}N(0,I(\theta)^{-1})$.

The proof follows Theorem 6.13, expanding $\dot l_n(\hat\theta_n^{(0)})$ around $\theta$.

**Example 6.15 (Logistic location)**

Start from $\bar X$, with $\sqrt{n}(\bar X-\theta)\xrightarrow{d}N(0,\pi^2/3)$. The information is

$$I(\theta)=E_\theta\big[-\partial_\theta^2\log f(X;\theta)\big]=2E\Big[\frac{e^{-(X-\theta)}}{(1+e^{-(X-\theta)})^2}\Big]=2E[U(1-U)]=\frac13,$$

as $U=1/(1+e^{-(X-\theta)})\sim U(0,1)$. So $\sqrt{n}(\hat\theta^{(1)}-\theta)\xrightarrow{d}N(0,3)$, better than $\pi^2/3\approx3.29$.

**Example 6.16 (Gamma)**

Gamma$(\alpha,\beta)$, $\Psi=(\log\Gamma)'$:

$$\dot l_n=\begin{pmatrix}-n\Psi(\alpha)-n\log\beta+n\overline{\log x}\\ -n\alpha/\beta+n\bar x/\beta^2\end{pmatrix},\qquad I(\theta)=\begin{pmatrix}\dot\Psi(\alpha)&1/\beta\\1/\beta&\alpha/\beta^2\end{pmatrix}.$$

From the $\sqrt{n}$-consistent MME (Example 6.1(d), Theorem 6.2), one Newton step gives the limit $N_2(\mathbf{0},I(\theta)^{-1})$,

$$I(\theta)^{-1}=\frac{1}{\alpha\dot\Psi(\alpha)-1}\begin{pmatrix}\alpha&-\beta\\-\beta&\beta^2\dot\Psi(\alpha)\end{pmatrix}.$$

## 6.5 Least squares

In the **linear regression model**

$$Y_i=x_{i0}\beta_0+x_{i1}\beta_1+\cdots+x_{ip}\beta_p+e_i,\qquad Ee_i=0,\ \operatorname{Var}e_i=\sigma^2,\ \operatorname{Cov}(e_i,e_j)=0\ (i\neq j),$$

the **least squares estimator** (LSE) $\hat\beta$ minimizes $\sum_i\{Y_i-(x_{i0}\beta_0+\cdots+x_{ip}\beta_p)\}^2$. In matrix form $\mathbf{Y}=X\beta+\mathbf{e}$, $X$ of size $n\times(p+1)$ and rank $p+1$, $E\mathbf{e}=\mathbf{0}$, $\operatorname{Var}\mathbf{e}=\sigma^2I_n$, and

$$\hat\beta^{\mathrm{LSE}}=\operatorname*{arg\,min}_{\beta\in\mathbb{R}^{p+1}}\|\mathbf{Y}-X\beta\|^2.$$

**Theorem 6.15 (Least squares)**

(a) $\Pi=X(X'X)^{-1}X'$ satisfies $\Pi'=\Pi$, $\Pi'(I-\Pi)=0$ and $\Pi X=X$.

(b) $\|\mathbf{Y}-X\beta\|^2=\|\Pi(\mathbf{Y}-X\beta)\|^2+\|(I-\Pi)\mathbf{Y}\|^2$; hence $X\hat\beta^{\mathrm{LSE}}=\Pi\mathbf{Y}$ and $\hat\beta^{\mathrm{LSE}}=(X'X)^{-1}X'\mathbf{Y}$.

**Proof.** (a) $\Pi'=X\{(X'X)^{-1}\}'X'=\Pi$ and

$$\Pi'\Pi=X(X'X)^{-1}X'X(X'X)^{-1}X'=\Pi,$$

so $\Pi'(I-\Pi)=0$, and $\Pi X = X$.

(b) $\mathbf{Y} - X\beta = \Pi(\mathbf{Y}-X\beta) + (I-\Pi)\mathbf{Y}$ with vanishing cross term; the second part is free of $\beta$, and the first is zero iff $X\beta = \Pi\mathbf{Y}$, i.e. $\beta = (X'X)^{-1}X'\mathbf{Y}$. ■

$\Pi$ projects orthogonally onto $\mathrm{col}(X)$. The error variance is estimated by

$$\hat\sigma^2 = \frac{\|\mathbf{Y} - X\hat\beta\|^2}{n-p-1}.$$

**Theorem 6.16 (Properties of the LSE)**

(a) $E\hat\beta = \beta$ and $\mathrm{Var}\,\hat\beta = \sigma^2(X'X)^{-1}$.

(b) $E\hat\sigma^2 = \sigma^2$.

(c) If $\mathbf{e} \sim N_n(\mathbf{0}, \sigma^2 I)$, then $\hat\beta \sim N(\beta, \sigma^2(X'X)^{-1})$, $(n-p-1)\hat\sigma^2/\sigma^2 \sim \chi^2(n-p-1)$, and they are independent.

**Proof.** (a) $E\hat\beta = (X'X)^{-1}X'X\beta = \beta$ and $\mathrm{Var}\,\hat\beta = (X'X)^{-1}X'(\sigma^2 I)X(X'X)^{-1} = \sigma^2(X'X)^{-1}$.

(b) $(I-\Pi)\mathbf{Y} = (I-\Pi)\mathbf{e}$, so $(n-p-1)\hat\sigma^2 = \mathbf{e}'(I-\Pi)\mathbf{e}$ and

$$E\,\mathbf{e}'(I-\Pi)\mathbf{e} = E\mathrm{tr}\{(I-\Pi)\mathbf{e}\mathbf{e}'\} = \mathrm{tr}\{(I-\Pi)\sigma^2 I\} = \sigma^2\{n - \mathrm{tr}((X'X)^{-1}X'X)\} = \sigma^2(n-p-1).$$

(c) is Theorem 4.19 in Volume I. ■

**Orthogonalization.** Partition $X = (X_0, X_1)$ ($p_0$ and $p_1$ columns), $\Pi_0 = X_0(X_0'X_0)^{-1}X_0'$. Then $X_{1|0} = (I-\Pi_0)X_1$ is orthogonal to $X_0$, and

$$X\beta = X_0\beta_0 + X_1\beta_1 = X_0\gamma_0 + X_{1|0}\beta_1, \qquad \gamma_0 = \beta_0 + (X_0'X_0)^{-1}X_0'X_1\beta_1.$$

With $\Pi_{1|0} = X_{1|0}(X_{1|0}'X_{1|0})^{-1}X_{1|0}'$, $\Pi = \Pi_0 + \Pi_{1|0}$ and $\Pi_0\Pi_{1|0} = 0$.

**Theorem 6.17**

In this notation,

(a) $X_0\hat\gamma_0 = \Pi_0\mathbf{Y}$, $X_{1|0}\hat\beta_1 = \Pi_{1|0}\mathbf{Y}$, i.e. $\hat\gamma_0 = (X_0'X_0)^{-1}X_0'\mathbf{Y}$ and $\hat\beta_1 = (X_{1|0}'X_{1|0})^{-1}X_{1|0}'\mathbf{Y}$, and $\hat\beta_0 = \hat\gamma_0 - (X_0'X_0)^{-1}X_0'X_1\hat\beta_1$;

(b) $\mathrm{Cov}(\hat\gamma_0, \hat\beta_1) = 0$, $\mathrm{Var}\,\hat\gamma_0 = \sigma^2(X_0'X_0)^{-1}$ and $\mathrm{Var}\,\hat\beta_1 = \sigma^2(X_{1|0}'X_{1|0})^{-1}$.

**Proof.** (a) The design $(X_0, X_{1|0})$ has block-diagonal Gram matrix, so $(X'X)^{-1}X'\mathbf{Y}$ splits; invert the definition of $\gamma_0$ for $\hat\beta_0$.

(b) $\mathrm{Cov}(\hat\gamma_0, \hat\beta_1) = \sigma^2(X_0'X_0)^{-1}X_0'X_{1|0}(X_{1|0}'X_{1|0})^{-1} = 0$; variances by Theorem 6.16(a). ■

**Example 6.17 (Centring the explanatory variables)**

$EY_i = \gamma_0 + \sum_j \beta_j(x_{ij} - \bar x_{\cdot j})$, $\gamma_0 = \beta_0 + \sum_j \beta_j \bar x_{\cdot j}$: $X_0 = \mathbf{1}$, $X_{1|0}$ has entries $x_{ij} - \bar x_{\cdot j}$. With $S_{jk} = \sum_i (x_{ij} - \bar x_{\cdot j})(x_{ik} - \bar x_{\cdot k})$, $S_{jY} = \sum_i (x_{ij} - \bar x_{\cdot j})(Y_i - \bar Y)$,

$$\hat\gamma_0 = \bar Y, \qquad \begin{pmatrix} S_{11} & \cdots & S_{1p} \\ \vdots & & \vdots \\ S_{p1} & \cdots & S_{pp} \end{pmatrix} \begin{pmatrix} \hat\beta_1 \\ \vdots \\ \hat\beta_p \end{pmatrix} = \begin{pmatrix} S_{1Y} \\ \vdots \\ S_{pY} \end{pmatrix}.$$

For $p = 1$: $\hat{\beta}_1 = S_{1Y}/S_{11}$, $\hat{\beta}_0 = \bar{Y} - \hat{\beta}_1 \bar{x}$, $\operatorname{Var} \hat{\beta}_1 = \sigma^2/S_{11}$, $\operatorname{Cov}(\bar{Y}, \hat{\beta}_1) = 0$.

# Chapter 7. Hypothesis Testing

## 7.1 The logic of testing

If the observed result would be very unlikely under a proposed model, we reject the model in favour of an alternative: **statistical hypothesis testing**.

**Example 7.1**

A standard painkiller acts in $N(20, 4^2)$ minutes; a new drug is claimed faster. With $n = 64$, $\bar{X} \sim N(\mu, 0.5^2)$; test $H_1 : \mu < 20$ against $H_0 : \mu = 20$.
Small $\bar{X}$ is hard to explain under $H_0$:

$$P(\bar{X} \le 19 \mid H_0) = P\Big(\frac{\bar{X} - 20}{0.5} \le -2\Big) = 0.023 :$$

about twice in a hundred trials under $H_0$: strong evidence against it.

$H_0$ is the **null hypothesis**, $H_1$ the **alternative**. The set $C$ of samples rejecting $H_0$ is the **rejection** (critical) region, its complement the **acceptance region**; a statistic describing $C$ (here $\bar{X}$) is a **test statistic**. Errors:

| | $H_0$ true | $H_1$ true |
|---|---|---|
| accept $H_0$ | correct | type II error |
| reject $H_0$ | type I error | correct |

We bound the type I error probability by a small **significance level** $\alpha$ (0.10, 0.05, 0.01): $P(\text{reject} \mid H_0) \le \alpha$ defines a **level-$\alpha$ test**. To minimize type II error, make the rejection region as large as the level allows.

**Example 7.2**

"$\bar{X} \le 19$" (probability 0.023 under $H_0$) and "$\bar{X} \le 20 - z_{0.05} \cdot 0.5 = 19.1775$" (exactly 0.05) are both level 0.05. Under $\mu$,

$$P_\mu(\bar{X} \le 19) = \Phi(38 - 2\mu), \qquad P_\mu(\bar{X} \le 19.1775) = \Phi(38.355 - 2\mu).$$

The second is larger for every $\mu < 20$ (smaller type II error); generally the region is $\bar{X} \le 20 - z_\alpha \cdot 0.5$.

**Example 7.3 (Randomization)**

Poisson($\theta$) sample, $n = 50$; test $H_0 : \theta = 0.1$ vs $H_1 : \theta < 0.1$ at $\alpha = 0.05$. Small $S = \sum X_i$ speaks against $H_0$, and $S \sim$ Poisson(5) under $H_0$:

$$P_{0.1}(S \le 0) = 0.0067, \qquad P_{0.1}(S \le 1) = 0.0404, \qquad P_{0.1}(S \le 2) = 0.1247.$$

No region $S \le k$ has probability exactly 0.05; reject for $S \le 1$, with probability $\gamma$ for $S = 2$, accept for $S \ge 3$:

$$0.0404 + \gamma \cdot P_{0.1}(S = 2) = 0.05, \qquad \gamma = \frac{0.05 - 0.0404}{0.0842} = 0.114.$$

a **randomized test**.

**General formulation.** Let the model be $\{f(x;\theta):\theta\in\Omega\}$ and test

$$H_0:\theta\in\Omega_0 \quad \text{vs} \quad H_1:\theta\in\Omega_1, \qquad \Omega_0\cap\Omega_1=\varnothing,\ \Omega_0\cup\Omega_1=\Omega.$$

A (possibly randomized) test is a **test function** $\phi(\mathbf{x})\in[0,1]$, the probability of rejecting at $\mathbf{X}=\mathbf{x}$ ($\phi=\mathbf{1}_C$ if nonrandomized). Its **power function** is $\gamma_\phi(\theta)=E_\theta\phi(\mathbf{X})$, its **size** $\max_{\theta\in\Omega_0}E_\theta\phi(\mathbf{X})$; level $\alpha$ means size $\le\alpha$. We seek size-$\alpha$ tests with maximal power on $\Omega_1$.

**Example 7.4 (Composite null)**

In Example 7.1 with $H_0:\mu\ge 20$: $P_\mu(\bar X\le c)=\Phi((c-\mu)/0.5)$ decreases in $\mu$, so the size is attained at $\mu=20$, giving the same region $\bar X\le 20-0.5z_\alpha$, with power $\gamma(\mu)=\Phi(38.355-2\mu)$ ($\alpha=0.05$). In Example 7.3

$$\gamma(\theta)=\sum_{x=0}^{1}e^{-50\theta}\frac{(50\theta)^x}{x!}+0.114\,e^{-50\theta}\frac{(50\theta)^2}{2},$$

also decreasing in $\theta$.

## 7.2 Likelihood ratio tests

Compare the best fits under each hypothesis:

$$\frac{\max_{\theta\in\Omega_1}L(\theta;\mathbf{x})}{\max_{\theta\in\Omega_0}L(\theta;\mathbf{x})}.$$

Large values are evidence against $H_0$. Since $\max_\Omega L=\max\{\max_{\Omega_0}L,\max_{\Omega_1}L\}$,

$$\frac{\max_\Omega L}{\max_{\Omega_0}L}=\max\Big\{1,\frac{\max_{\Omega_1}L}{\max_{\Omega_0}L}\Big\},$$

which is monotone in the first ratio; so use the full maximum, with MLEs $\hat\theta^{\Omega}$, $\hat\theta^{\Omega_0}$:

**Definition (Likelihood ratio test)**

The **likelihood ratio test** (LRT) of level $\alpha$ rejects $H_0$ when

$$2\{l(\hat\theta^{\Omega})-l(\hat\theta^{\Omega_0})\}\ge c, \qquad \text{with } c \text{ chosen so that} \quad \max_{\theta\in\Omega_0}P_\theta(\text{reject})=\alpha$$

(randomizing if necessary).

**Example 7.5 (Two-sided $t$ test)**

$N(\mu,\sigma^2)$, $n\ge 2$; $H_0:\mu=\mu_0$ vs $H_1:\mu\ne\mu_0$. With

$$l(\mu,\sigma^2)=-\frac{1}{2\sigma^2}\sum(x_i-\mu)^2-\frac{n}{2}\log\sigma^2-\frac{n}{2}\log 2\pi,$$

$\hat\theta^{\Omega}=(\bar x,\hat\sigma^2)$, $\hat\sigma^2=\frac{1}{n}\sum(x_i-\bar x)^2$, and $\hat\theta^{\Omega_0}=(\mu_0,\hat\sigma_0^2)$, $\hat\sigma_0^2=\hat\sigma^2+(\bar x-\mu_0)^2$. Hence

$$2\{l(\hat\theta^{\Omega})-l(\hat\theta^{\Omega_0})\}=n\log\frac{\hat\sigma_0^2}{\hat\sigma^2}=n\log\Big(1+\frac{(\bar x-\mu_0)^2}{\hat\sigma^2}\Big),$$

increasing in $|\bar{x}-\mu_0|/(s/\sqrt{n})$. Since $\sqrt{n}(\bar{X}-\mu_0)/S \sim t(n-1)$ under $H_0$, the LRT is the **two-sided $t$ test**:

$$\text{reject } H_0 \iff \Big|\frac{\bar{X}-\mu_0}{S/\sqrt{n}}\Big| \geq t_{\alpha/2}(n-1).$$

**Example 7.6 (One-sided $t$ test)**

$H_0: \mu \leq \mu_0$ vs $H_1: \mu > \mu_0$. Over $\Omega_0$, $\hat{\mu}^{\Omega_0} = \min(\bar{x}, \mu_0)$ and $\hat{\sigma}^{2,\Omega_0} = \frac{1}{n}\sum(x_i - \hat{\mu}^{\Omega_0})^2$, so

$$2\{l(\hat{\theta}^{\Omega}) - l(\hat{\theta}^{\Omega_0})\} = n\log\Big(1 + \frac{(\bar{x}-\min(\bar{x},\mu_0))^2}{\hat{\sigma}^2}\Big),$$

0 for $\bar{x} \leq \mu_0$, increasing in $(\bar{x}-\mu_0)/(s/\sqrt{n})$ otherwise: the region is $\{\sqrt{n}(\bar{x}-\mu_0)/s \geq c\}$, $c > 0$. With independent $Z = \sqrt{n}(\bar{X}-\mu)/\sigma \sim N(0,1)$, $V = (n-1)S^2/\sigma^2 \sim \chi^2(n-1)$,

$$P_{\mu,\sigma^2}\Big(\frac{\bar{X}-\mu_0}{S/\sqrt{n}} \geq c\Big) = P\Big(c\sqrt{V/(n-1)} - Z \leq \frac{\mu-\mu_0}{\sigma/\sqrt{n}}\Big),$$

increasing in $\mu$, so the size is at $\mu = \mu_0$: the **one-sided $t$ test** rejects when $\sqrt{n}(\bar{X}-\mu_0)/S \geq t_\alpha(n-1)$.

**Example 7.7 (Exponential mean)**

$\text{Exp}(\theta)$, $H_0: \theta = \theta_0$ vs $H_1: \theta \neq \theta_0$: $\hat{\theta}^{\Omega} = \bar{x}$ and

$$2\{l(\hat{\theta}^{\Omega}) - l(\theta_0)\} = 2n\Big(\frac{\bar{x}}{\theta_0} - 1 - \log\frac{\bar{x}}{\theta_0}\Big).$$

As $t - 1 - \log t$ decreases on $(0,1)$ and increases on $(1,\infty)$, the region is

$$\frac{\bar{x}}{\theta_0} \leq c_1 \quad \text{or} \quad \frac{\bar{x}}{\theta_0} \geq c_2, \qquad c_1 - \log c_1 = c_2 - \log c_2.$$

Under $H_0$, $2n\bar{X}/\theta_0 \sim \chi^2(2n)$, so $\int_{2nc_1}^{2nc_2} f_{\chi^2(2n)} = 1-\alpha$ determines $c_1, c_2$. For large $n$, $c_{1,2} \approx 1 \mp z_{\alpha/2}/\sqrt{n}$: approximately "$n(\bar{X}/\theta_0 - 1)^2 \geq \chi^2_\alpha(1)$".

**Example 7.8 (Two-sample $t$ test)**

Independent $N(\mu_i, \sigma^2)$ samples, $i = 1, 2$; $H_0: \mu_1 = \mu_2$. Under $\Omega$, $\hat{\mu}_i = \bar{x}_i$, $\hat{\sigma}^2 = \sum_{i,j}(x_{ij} - \bar{x}_i)^2/(n_1+n_2)$; under $\Omega_0$, $\hat{\mu} = (n_1\bar{x}_1 + n_2\bar{x}_2)/(n_1+n_2)$ and $\hat{\sigma}_0^2 = \hat{\sigma}^2 + \frac{n_1 n_2}{(n_1+n_2)^2}(\bar{x}_1 - \bar{x}_2)^2$:

$$2\{l(\hat{\theta}^{\Omega}) - l(\hat{\theta}^{\Omega_0})\} = (n_1+n_2)\log\Big(1 + \frac{n_1 n_2}{(n_1+n_2)^2}\frac{(\bar{x}_1-\bar{x}_2)^2}{\hat{\sigma}^2}\Big),$$

increasing in $|\bar{x}_1 - \bar{x}_2|/(s_p\sqrt{1/n_1 + 1/n_2})$, $s_p^2 = \sum_{i,j}(x_{ij} - \bar{x}_i)^2/(n_1+n_2-2)$, which is $t(n_1 + n_2 - 2)$ under $H_0$ (Corollary 4.10 in Volume I). Reject when

$$\frac{|\bar{X}_1 - \bar{X}_2|}{S_p\sqrt{1/n_1 + 1/n_2}} \geq t_{\alpha/2}(n_1+n_2-2).$$

**Example 7.9 (Comparing two variances)**

$N(\mu_i, \sigma_i^2)$ samples; $H_0 : \sigma_1^2 = \sigma_2^2$. With $\hat\sigma_i^2 = \frac{1}{n_i}\sum_j (x_{ij} - \bar x_i)^2$ and pooled $\hat\sigma^2 = (n_1\hat\sigma_1^2 + n_2\hat\sigma_2^2)/(n_1+n_2)$,

$$2\{l(\hat\theta^{\Omega}) - l(\hat\theta^{\Omega_0})\} = (n_1+n_2)\log\hat\sigma^2 - n_1\log\hat\sigma_1^2 - n_2\log\hat\sigma_2^2.$$

With $t = n_1\hat\sigma_1^2/(n_1\hat\sigma_1^2 + n_2\hat\sigma_2^2)$ this is $-n_1\log t - n_2\log(1-t) + \text{const}$, minimal at $t = n_1/(n_1+n_2)$; $t$ increases in $F = S_1^2/S_2^2 \sim F(n_1-1, n_2-1)$ under $H_0$, so the region is $F \le d_1$ or $F \ge d_2$ ($\int_{d_1}^{d_2} f_F = 1-\alpha$, equal statistic at $d_1, d_2$). In practice $d_1 = F_{1-\alpha/2}$, $d_2 = F_{\alpha/2}$.

**Example 7.10 (Poisson, one-sided)**

As in Example 7.3, $H_0 : \theta \ge 0.1$ vs $H_1 : \theta < 0.1$: $\hat\theta^{\Omega} = \bar x$, $\hat\theta^{\Omega_0} = \max(\bar x, 0.1)$, and

$$2\{l(\hat\theta^{\Omega}) - l(\hat\theta^{\Omega_0})\} = 2n\{\bar x\log\bar x - \bar x(1+\log 0.1) + 0.1\}\mathbf{1}\{\bar x < 0.1\},$$

decreasing in $\bar x \le 0.1$: reject for small $S$. As $P_\theta(S \le k) = \int_{n\theta}^{\infty} z^k e^{-z}/k!\,dz$ decreases in $\theta$, the size is at $\theta = 0.1$, and Example 7.3 is the level-0.05 LRT.

## 7.3 Large-sample approximations

Without exact null laws, use MLE asymptotics: $\Omega \subseteq \mathbb{R}^k$ open, $\bar l = l/n$, Fisher information $I(\theta)$, and $\sqrt{n}(\hat\theta_n - \theta) \approx I(\theta)^{-1}\sqrt{n}\,\dot{\bar l}(\theta)$ (Chapter 6).

**Simple null hypothesis.** $H_0 : \theta = \theta_0$. Expanding about $\hat\theta_n$ ($\dot l(\hat\theta_n) = 0$),

$$l(\theta_0) = l(\hat\theta_n) + \tfrac{1}{2}(\theta_0 - \hat\theta_n)'\ddot l(\theta_n^*)(\theta_0 - \hat\theta_n),$$

with $-\frac{1}{n}\ddot l(\theta_n^*) \xrightarrow{p} I(\theta_0)$ under $H_0$. Hence

$$2\{l(\hat\theta_n) - l(\theta_0)\} \approx \sqrt{n}(\hat\theta_n - \theta_0)'I(\theta_0)\sqrt{n}(\hat\theta_n - \theta_0).$$

and $\sqrt{n}(\hat\theta_n - \theta_0) \xrightarrow{d} \mathbf{Z} \sim N_k(\mathbf{0}, I(\theta_0)^{-1})$ gives the limit $\mathbf{Z}'I(\theta_0)\mathbf{Z} \sim \chi^2(k)$ (Theorem 4.18 in Volume I).

**Theorem 7.1 (Simple null hypothesis)**

Under (R0)–(R7) of Theorem 6.13, for $H_0 : \theta = \theta_0$ vs $H_1 : \theta \ne \theta_0$, under $H_0$:

(a) $2\{l(\hat\theta_n) - l(\theta_0)\} \xrightarrow{d} \chi^2(k)$; rejecting when it is $\ge \chi^2_\alpha(k)$ has asymptotic level $\alpha$.

(b) $2\{l(\hat\theta_n) - l(\theta_0)\} = W_n(\theta_0) + o_p(1)$, where $W_n(\theta_0) = n(\hat\theta_n - \theta_0)'I(\theta_0)(\hat\theta_n - \theta_0)$.

(c) $2\{l(\hat\theta_n) - l(\theta_0)\} = R_n(\theta_0) + o_p(1)$, where $R_n(\theta_0) = n\,\dot{\bar l}(\theta_0)'I(\theta_0)^{-1}\dot{\bar l}(\theta_0)$.

**Sketch.** (a), (b) as above (remainder as in Theorem 6.13). (c): substitute $\sqrt{n}(\hat\theta_n - \theta_0) = I(\theta_0)^{-1}\sqrt{n}\,\dot{\bar l}(\theta_0) + o_p(1)$ into $W_n$. ■

$W_n$ is **Wald's statistic**, $R_n$ **Rao's (score) statistic**; all three are asymptotically $\chi^2(k)$. The score test needs only $\theta_0$.

**Example 7.11 (Exponential)**

$\mathrm{Exp}(\theta)$, $H_0:\theta=\theta_0$: $\dot l(\theta)=-\frac{1}{\theta}+\frac{\bar x}{\theta^2}$, $\hat\theta=\bar x$, $I(\theta)=1/\theta^2$:

$$2\{l(\hat\theta)-l(\theta_0)\}=2n\Big(\frac{\bar x}{\theta_0}-1-\log\frac{\bar x}{\theta_0}\Big),\qquad W_n=\frac{n(\bar x-\theta_0)^2}{\bar x^2},\qquad R_n=\frac{n(\bar x-\theta_0)^2}{\theta_0^2}.$$

each compared with $\chi^2_\alpha(1)$.

**Example 7.12 (Multinomial goodness of fit)**

$\mathbf{Z}_i$ i.i.d. Multi$(1,\mathbf{p})$, counts $X_j=\sum_i Z_{ij}$, $H_0:\mathbf{p}=\mathbf{p}_0$. With $\theta=(p_1,\dots,p_r)'$, $r=k-1$, $\theta_\cdot=\sum_{j\le r}\theta_j=1-p_k$,

$$l(\theta)=\sum_{j\le r}x_j\log\theta_j+x_k\log(1-\theta_\cdot),\qquad \hat\theta_j=x_j/n,$$

$$I(\theta)=\mathrm{diag}(1/\theta_j)+\frac{1}{p_k}\mathbf{1}\mathbf{1}'.$$

One computes $I(\theta_0)(\hat\theta-\theta_0)=\dot l(\theta_0)$, with $j$-th entry $\frac{x_j-np_{0j}}{np_{0j}}-\frac{x_k-np_{0k}}{np_{0k}}$, so

$$W_n(\theta_0)=R_n(\theta_0)=\sum_{j=1}^{k}\frac{(X_j-np_{0j})^2}{np_{0j}},$$

Pearson's chi-square; the LRT statistic is $2\sum_j X_j\log\frac{X_j}{np_{0j}}$. All are asymptotically $\chi^2(k-1)$ (cf. Example 5.12 in Volume I).

**Composite null hypotheses.** Write $\theta=(\xi',\eta')'$, $\xi\in\mathbb{R}^{k_1}$, $\eta\in\mathbb{R}^{k_0}$, $k_0+k_1=k$, $H_0:\xi=\xi_0$, and partition

$$\dot l=\begin{pmatrix}\dot l_1\\ \dot l_2\end{pmatrix},\qquad I=\begin{pmatrix}I_{11}&I_{12}\\ I_{21}&I_{22}\end{pmatrix},\qquad I_{11\cdot2}=I_{11}-I_{12}I_{22}^{-1}I_{21}.$$

with $\hat\theta_n$ the unrestricted MLE and $\hat\theta_n^0=(\xi_0',\hat\eta_n^{0\prime})'$ the MLE under $H_0$.

**Theorem 7.2 (Composite null hypothesis)**

Under Theorem 6.13's conditions, if $H_0:\xi=\xi_0$ holds with $\theta_0=(\xi_0',\eta_0')'$,

(a) $2\{l(\hat\theta_n)-l(\hat\theta_n^0)\}\xrightarrow{d}\chi^2(k-k_0)$;

(b) $2\{l(\hat\theta_n)-l(\hat\theta_n^0)\}=n(\hat\theta_n-\hat\theta_n^0)'I(\theta_0)(\hat\theta_n-\hat\theta_n^0)+o_p(1)$;

(c) $2\{l(\hat\theta_n)-l(\hat\theta_n^0)\}=n\,\dot l(\hat\theta_n^0)'I(\theta_0)^{-1}\dot l(\hat\theta_n^0)+o_p(1)$.

**Sketch.** As in Theorem 7.1,

$$2\{l(\hat\theta_n)-l(\theta_0)\}\approx\sqrt{n}\,\dot l(\theta_0)'I^{-1}\sqrt{n}\,\dot l(\theta_0).$$

and for the submodel with $\xi=\xi_0$ fixed,

$$2\{l(\hat\theta_n^0)-l(\theta_0)\}\approx\sqrt{n}\,\dot l_2(\theta_0)'I_{22}^{-1}\sqrt{n}\,\dot l_2(\theta_0).$$

Subtracting and using

$$\begin{pmatrix} I_{11} & I_{12} \\ I_{21} & I_{22} \end{pmatrix}^{-1} = \begin{pmatrix} I_{11\cdot 2}^{-1} & -I_{11\cdot 2}^{-1} I_{12} I_{22}^{-1} \\ -I_{22}^{-1} I_{21} I_{11\cdot 2}^{-1} & I_{22}^{-1} + I_{22}^{-1} I_{21} I_{11\cdot 2}^{-1} I_{12} I_{22}^{-1} \end{pmatrix},$$

gives

$$2\{l(\hat\theta_n) - l(\hat\theta_n^0)\} \approx \sqrt{n}\,\mathbf{Z}_n' I_{11\cdot 2}^{-1} \sqrt{n}\,\mathbf{Z}_n, \qquad \mathbf{Z}_n = \dot l_1(\theta_0) - I_{12} I_{22}^{-1} \dot l_2(\theta_0).$$

By the CLT $\sqrt{n}\,\mathbf{Z}_n \xrightarrow{d} N_{k_1}(\mathbf{0}, I_{11\cdot 2})$, so the limit is $\chi^2(k_1)$: (a). For (b), $\hat\eta_n^0 - \eta_0 \approx I_{22}^{-1}\dot l_2(\theta_0)$ and $I_{11\cdot 2}(\hat\xi_n - \xi_0) \approx \mathbf{Z}_n$ give $n(\hat\theta_n - \hat\theta_n^0)' I (\hat\theta_n - \hat\theta_n^0) \approx n\mathbf{Z}_n' I_{11\cdot 2}^{-1} \mathbf{Z}_n$;

(c) is similar. ■

With $I(\hat\theta_n^0)$ for $I(\theta_0)$, (b) and (c) are the composite Wald and Rao statistics. Smooth constraints $H_0$ : $g_1(\theta) = \cdots = g_{k_1}(\theta) = 0$ are handled by reparametrizing $\xi_j = g_j(\theta)$; the limit is $\chi^2(\dim\Omega - \dim\Omega_0)$.

**Example 7.13 (Independence in a contingency table)**

Counts $X_{ij}$ of an $r \times c$ table are Multi$(n, (p_{ij}))$; $H_0 : p_{ij} = p_{i\cdot}p_{\cdot j}$. MLEs: $\hat p_{ij} = x_{ij}/n$, and $\hat p_{ij}^0 = \frac{x_{i\cdot}}{n}\frac{x_{\cdot j}}{n}$ under $H_0$:

$$2\{l(\hat p) - l(\hat p^0)\} = 2\sum_{i,j} X_{ij} \log \frac{X_{ij}}{n\hat p_{i\cdot}\hat p_{\cdot j}}.$$

$\dim\Omega = rc - 1$, $\dim\Omega_0 = (r-1) + (c-1)$, so the limit is $\chi^2((r-1)(c-1))$. Pearson's statistic with estimated marginals gives the **chi-square test of independence**:

$$\sum_{i=1}^{r}\sum_{j=1}^{c} \frac{(O_{ij} - \hat E_{ij})^2}{\hat E_{ij}} \geq \chi_\alpha^2((r-1)(c-1)), \qquad O_{ij} = X_{ij},\ \hat E_{ij} = \frac{X_{i\cdot}X_{\cdot j}}{n}.$$

**Example 7.14 (A worked $2 \times 3$ table)**

200 students by year and preferred study place:

| | library | home | cafe | total |
|---|---|---|---|---|
| lower | 40 | 30 | 30 | 100 |
| upper | 20 | 50 | 30 | 100 |
| total | 60 | 80 | 60 | 200 |

$\hat E_{1j} = \hat E_{2j} = (30, 40, 30)$ and

$$2\Big(\frac{10^2}{30} + \frac{10^2}{40} + 0\Big) = 11.67 > \chi_{0.05}^2(2) = 5.99,$$

reject independence at 5%.

**Non-identically distributed observations.** Under suitable conditions the results hold with $\bar I_n(\theta) = \frac{1}{n}E_\theta[-\ddot l(\theta)]$ for $I(\theta)$: $W_n = n(\hat\theta - \theta_0)'\bar I_n(\theta_0)(\hat\theta - \theta_0)$, $R_n = n\,\dot l(\theta_0)'\bar I_n(\theta_0)^{-1}\dot l(\theta_0)$.

**Example 7.15 (Homogeneity of multinomial populations)**

Independent $\mathbf{X}_i \sim$ Multi$(n_i, \mathbf{p}_i)$, $i \leq r$, $c$ cells; $H_0 : \mathbf{p}_1 = \cdots = \mathbf{p}_r$, pooled $\hat p_j^0 = X_{\cdot j}/n$. With $O_{ij} = X_{ij}$, $\hat E_{ij} = n_i \hat p_j^0$, the LRT statistic $2\sum O_{ij}\log(O_{ij}/\hat E_{ij})$ is approximately Pearson's $\sum\sum (O_{ij} - \hat E_{ij})^2/\hat E_{ij}$ ($2u\log u \approx 2(u-1) + (u-1)^2$ near $u = 1$), with $r(c-1) - (c-1) =$

$(r-1)(c-1)$ d.f., as for independence though row totals are fixed. (The limit follows from a quadratic form in $\sqrt{n_i}(\hat{\mathbf{p}}_i - \hat{\mathbf{p}}^0)$ with an idempotent matrix of rank $(r-1)(c-1)$.)

## 7.4 Selected exercises

**Motivation.** Book exercises: new LRTs, power of the $t$ test, and the Poisson homogeneity statistic as Pearson's $\chi^2$.

**Exercise 7.6 (Power of the two-sided $t$ test)**

Let $X_1, \dots, X_n$ $(n \geq 2)$ be $N(\mu, \sigma^2)$ and consider the two-sided $t$ test of Example 7.5 with statistic $t = (\bar{X} - \mu_0)/(S/\sqrt{n})$.
(a) Show that $t \overset{d}{=} (Z + \delta)/\sqrt{V/(n-1)}$, where $\delta = (\mu - \mu_0)/(\sigma/\sqrt{n})$, $Z \sim N(0,1)$, $V \sim \chi^2(n-1)$, and $Z, V$ are independent.
(b) Show that the power $P_{\mu,\sigma^2}\{|t| \geq t_{\alpha/2}(n-1)\}$ is a function of $|\delta|$ alone and is increasing in $|\delta|$.

**Solution.**

(a) With $Z = (\bar{X}-\mu)/(\sigma/\sqrt{n})$ and $V = (n-1)S^2/\sigma^2$, independent $N(0,1)$ and $\chi^2(n-1)$ (Chapter 4), dividing by $\sigma/\sqrt{n}$:

$$t = \frac{(\bar{X} - \mu)/(\sigma/\sqrt{n}) + (\mu - \mu_0)/(\sigma/\sqrt{n})}{S/\sigma} = \frac{Z + \delta}{\sqrt{V/(n-1)}}.$$

(b)

*Step 1.* With $c = t_{\alpha/2}(n-1)$ and $g_k(\delta) = P(|Z + \delta| \geq k) = 1 - \Phi(k - \delta) + \Phi(-k - \delta)$, by independence

$$\gamma(\delta) = P(|t| \geq c) = E\big[g_K(\delta)\big], \qquad K = c\sqrt{V/(n-1)}.$$

*Step 2.* $-Z \overset{d}{=} Z$ gives $g_k(-\delta) = g_k(\delta)$, so $\gamma$ depends on $|\delta|$.

*Step 3.* $g_k'(\delta) = \phi(k - \delta) - \phi(k + \delta) > 0$ for $\delta > 0$ (as $|k - \delta| < k + \delta$).

*Step 4.* $K > 0$ a.s., so $g_K(\delta_1) < g_K(\delta_2)$ for $0 \leq \delta_1 < \delta_2$; take expectations. ($t$ is *noncentral* $t$ with noncentrality $\delta$.) ■

**Exercise 7.9 (A one-sided test for Beta$(\theta, 1)$)**

Let $X_1, \dots, X_n$ have density $f(x;\theta) = \theta x^{\theta-1}$ on $(0,1)$, $\theta > 0$. Find the level-$\alpha$ LRT of $H_0 : \theta \leq 1$ against $H_1 : \theta > 1$ $(0 < \alpha < \frac{1}{2})$.

**Solution.**

*Step 1.* With $T = -\sum_i \log X_i > 0$, $l(\theta) = n \log \theta - (\theta - 1)T$ is strictly concave, maximal at $\hat{\theta} = n/T$; over $(0, 1]$ the MLE is $\min(\hat{\theta}, 1)$.

*Step 2.* For $\hat{\theta} \leq 1$ the statistic is 0; for $\hat{\theta} > 1$ $(T < n)$

$$2\{l(\hat{\theta}) - l(1)\} = 2\{n \log(n/T) - n + T\} =: h(T), \qquad h'(T) = 2\Big(1 - \frac{n}{T}\Big) < 0.$$

$h$ decreases from $+\infty$ to 0 on $(0, n)$, so the LRT rejects for $T \leq d$, $d \in (0, n)$.

*Step 3.* $P(-\log X_i > s) = e^{-\theta s}$, so $2\theta T \sim \chi^2(2n)$ and $P_\theta(T \le d) = P\{\chi^2(2n) \le 2\theta d\}$ increases in $\theta$: size at $\theta = 1$.

*Step 4.* So reject iff

$$-2\sum_{i=1}^{n} \log X_i \le \chi^2_{1-\alpha}(2n),$$

the lower $\alpha$ point. As $\alpha < \frac{1}{2}$ it lies below the median of $\chi^2(2n)$, which is below the mean $2n$ (gamma median $<$ mean), so $d < n$ as needed. ■

**Exercise 7.15(a) (Testing the endpoint of a uniform law)**

Let $X_1, \dots, X_n$ be $U[0, \theta]$, $\theta > 0$. Find the level-$\alpha$ LRT of $H_0 : \theta = \theta_0$ against $H_1 : \theta \ne \theta_0$.

**Solution.**

*Step 1.* With $M = \max_i X_i$, $L(\theta) = \theta^{-n} I(\theta \ge M)$, $\hat\theta = M$, and

$$\frac{L(\theta_0)}{L(\hat\theta)} = \begin{cases} (M/\theta_0)^n, & M \le \theta_0, \\ 0, & M > \theta_0. \end{cases}$$

Reject when the ratio is small: $M > \theta_0$ or $M \le k\theta_0$, $k \in (0, 1)$.

*Step 2.* Under $H_0$, $P(M > \theta_0) = 0$, $P(M \le k\theta_0) = k^n = \alpha$:

$$\text{reject } H_0 \iff M > \theta_0 \ \text{ or } \ M \le \alpha^{1/n}\theta_0,$$

of size exactly $\alpha$. Unlike Section 7.3, $2\{l(\hat\theta) - l(\theta_0)\} = -2n\log(M/\theta_0)$ is exactly $\chi^2(2)$ ($P(M/\theta_0 \le e^{-s/n}) = e^{-s}$). ■

**Exercise 7.17(a)(b) (Homogeneity of Poisson means)**

Let $X_{i1}, \dots, X_{in_i}$ be a sample from Poisson$(\theta_i)$, $i = 1, \dots, k$ ($k \ge 2$), the samples being independent. Put $X_{i\cdot} = \sum_j X_{ij}$, $n = n_1 + \cdots + n_k$, $\hat\theta_i = X_{i\cdot}/n_i$ and $\hat\theta^0 = \sum_i X_{i\cdot}/n$. For $H_0 : \theta_1 = \cdots = \theta_k$:
(a) Show that the LRT statistic is $2\sum_i n_i\{\hat\theta_i \log(\hat\theta_i/\hat\theta^0) - \hat\theta_i + \hat\theta^0\}$ (with $0 \log 0 = 0$).
(b) Show that if $n_i \to \infty$ with $n_i/n \to \gamma_i \in (0, 1)$, then under $H_0$ (with common value $\theta > 0$) the statistic is approximately $\sum_i (X_{i\cdot} - n_i\hat\theta^0)^2/(n_i\hat\theta^0)$, in the sense that the ratio of the two tends to 1 in probability.

**Solution.**

(a) Up to a constant, $l(\theta) = \sum_i \{X_{i\cdot} \log\theta_i - n_i\theta_i\}$, maximized at $\hat\theta_i$, and under $H_0$ at $\hat\theta^0$. With $X_{i\cdot} = n_i\hat\theta_i$,

$$2\{l(\hat\theta) - l(\hat\theta^0)\} = 2\sum_i \{n_i\hat\theta_i \log(\hat\theta_i/\hat\theta^0) - n_i\hat\theta_i + n_i\hat\theta^0\}.$$

(The linear terms sum to 0.)

(b)

*Step 1.* For $a > 0$, $g_a(x) = x\log(x/a) - x + a$ has $g_a(a) = g_a'(a) = 0$, $g_a''(x) = 1/x$; by Taylor, for $x > 0$

$$g_a(x) = \frac{(x-a)^2}{2\xi}, \qquad \xi \text{ between } x \text{ and } a.$$

*Step 2.* Under $H_0$, $\hat\theta_i, \hat\theta^0 \xrightarrow{p} \theta$ (LLN), so $P(\hat\theta_i > 0) \to 1$, $\xi_i \xrightarrow{p} \theta$, and

$$2\{l(\hat\theta) - l(\hat\theta^0)\} = \sum_i n_i \frac{(\hat\theta_i - \hat\theta^0)^2}{\hat\theta^0} \cdot \frac{\hat\theta^0}{\xi_i},$$

with $\hat{\theta}^0/\xi_i \xrightarrow{p} 1$.

*Step 3.* $\sum_i a_i b_i / \sum_i a_i$ lies between $\min_i b_i$ and $\max_i b_i$ ($a_i \geq 0$), so it $\xrightarrow{p} 1$ for fixed $k$. With $n_i(\hat{\theta}_i - \hat{\theta}^0)^2/\hat{\theta}^0 = (X_{i\cdot} - n_i\hat{\theta}^0)^2/(n_i\hat{\theta}^0)$ this is the claim (limit $\chi^2(k-1)$, Theorem 7.2). ■

# Chapter 8. Comparing Estimators

## 8.1 Criteria for comparing estimators

The usual performance measure for estimating $\eta = \eta(\theta)$ is the **mean squared error**

$$MSE(\hat{\eta}, \theta) = E_\theta[\{\hat{\eta}(X_1, \ldots, X_n) - \eta(\theta)\}^2] = \operatorname{Var}_\theta \hat{\eta} + \{E_\theta \hat{\eta} - \eta(\theta)\}^2,$$

the variance plus the squared **bias**.

**Example 8.1 (A clear winner)**

$U(0, \theta)$, $n \geq 2$: $\hat{\theta}^{\mathrm{MLE}} = X_{(n)}$ vs $\hat{\theta}^{\mathrm{MME}} = 2\bar{X}$. With $X_{(n)}/\theta \sim \mathrm{Beta}(n, 1)$,

$$MSE(X_{(n)}, \theta) = \frac{2\theta^2}{(n+1)(n+2)}, \qquad MSE(2\bar{X}, \theta) = 4 \operatorname{Var} \bar{X} = \frac{\theta^2}{3n}.$$

The MLE is better for all $\theta$, $n$, with MSE of order $n^{-2}$ rather than $n^{-1}$.

**Example 8.2 (No uniform winner)**

$N(\theta, 1)$: $\bar{X}$ (MSE $1/n$) vs $\hat{\theta}^0 \equiv 0$ (MSE $\theta^2$), better for $|\theta| < 1/\sqrt{n}$; neither dominates. The "pretest" estimator

$$\hat{\theta}^H = \begin{cases} 0, & |\bar{X}| < 1.96/\sqrt{n}, \\ \bar{X}, & |\bar{X}| \geq 1.96/\sqrt{n}. \end{cases}$$

has MSE $n^{-1}E[(Z + \sqrt{n}\theta)^2 \mathbf{1}\{|Z + \sqrt{n}\theta| \geq 1.96\}] + \theta^2 P(|Z + \sqrt{n}\theta| < 1.96)$, $Z \sim N(0, 1)$, a function of $\sqrt{n}\theta$; $n \cdot MSE$ is below 1 near 0 but about 2.5 at $\sqrt{n}|\theta| \approx 2$, and $\to 1$ for fixed $\theta \neq 0$.

MSEs typically cross, so use criteria free of $\theta$:

- **Maximum MSE**: compare $\max_\theta MSE(\hat{\eta}, \theta)$; an estimator minimizing it is **minimax**.
- **Bayes MSE**: for a weight (prior) $\pi \geq 0$, $r(\pi, \hat{\eta}) = \int_\Omega MSE(\hat{\eta}, \theta)\pi(\theta)\, d\theta$; a minimizer is a **Bayes estimator**.

For positive targets, the relative MSE $E_\theta(\hat{\eta}/\eta - 1)^2$ is also used.

**Example 8.3**

$N(\mu, \sigma^2)$, $\hat{\sigma}_c^2 = c\sum(X_i - \bar{X})^2$; with $V = \sum(X_i - \bar{X})^2/\sigma^2 \sim \chi^2(n-1)$,

$$E_\theta(\hat{\sigma}_c^2/\sigma^2 - 1)^2 = E(cV - 1)^2 = c^2 \cdot 2(n-1) + \{c(n-1) - 1\}^2,$$

free of $\theta$ and minimized at $c = 1/(n+1)$: the minimax relative-MSE estimator is neither the MLE ($c = 1/n$) nor $S^2$.

Alternatively, restrict the class: $\hat{\eta}$ is **unbiased** if $E_\theta \hat{\eta} = \eta(\theta)$ for all $\theta$, and then $MSE = \operatorname{Var}_\theta \hat{\eta}$.

**Definition (UMVUE)**

$\hat{\eta}^*$ is a **uniformly minimum variance unbiased estimator** (UMVUE) of $\eta(\theta)$ if (i) it is unbiased, and (ii) $\operatorname{Var}_\theta \hat{\eta}^* \le \operatorname{Var}_\theta \hat{\eta}$ for all $\theta$ and every unbiased $\hat{\eta}$.

**Theorem 8.1 (Uniqueness)**

If the UMVUE of $\eta(\theta)$ has finite variance, it is unique: if $\hat{\eta}_1, \hat{\eta}_2$ are both UMVUEs, then $P_\theta(\hat{\eta}_1 = \hat{\eta}_2) = 1$ for all $\theta$.

**Proof.** $\hat{\eta}_3 = (\hat{\eta}_1 + \hat{\eta}_2)/2$ is unbiased, so $\operatorname{Var} \hat{\eta}_3 \ge \operatorname{Var} \hat{\eta}_2$:

$$\tfrac{1}{4}\{\operatorname{Var} \hat{\eta}_1 + 2\operatorname{Cov}(\hat{\eta}_1, \hat{\eta}_2) + \operatorname{Var} \hat{\eta}_2\} \ge \operatorname{Var} \hat{\eta}_2,$$

so $\operatorname{Cov}(\hat{\eta}_1, \hat{\eta}_2) \ge \operatorname{Var} \hat{\eta}_2$ and $\operatorname{Var}(\hat{\eta}_1 - \hat{\eta}_2) \le 0$; a zero-mean, zero-variance variable is 0 a.s. (Theorem 1.11 in Volume I). ■

## 8.2 Sufficient statistics

**Motivation.** Sufficiency captures all information about $\theta$ in a few numbers; factorization finds such statistics.

**Example 8.4**

Two Bernoulli($\theta$) trials, $Y = X_1 + X_2$. For $Y = 0, 2$ the data are determined; for $Y = 1$

$$P_\theta(X_1 = 0, X_2 = 1 \mid Y = 1) = \frac{(1-\theta)\theta}{2\theta(1-\theta)} = \frac{1}{2} = P_\theta(X_1 = 1, X_2 = 0 \mid Y = 1).$$

The data can be recreated from $Y$ by a coin flip without knowing $\theta$.

**Definition (Sufficiency)**

A statistic $Y = u(X_1, \ldots, X_n)$ is **sufficient** for $\theta \in \Omega$ if the conditional distribution of $(X_1, \ldots, X_n)$ given $Y = y$ does not depend on $\theta$, i.e. $P_{\theta_1}(\mathbf{X} \in A \mid Y = y) = P_{\theta_2}(\mathbf{X} \in A \mid Y = y)$ for all $A$, $y$, $\theta_1, \theta_2$.

**Example 8.5**

$n$ Bernoulli($\theta$) trials, $Y = \sum X_i \sim B(n, \theta)$:

$$P_\theta(X_1 = x_1, \ldots, X_n = x_n \mid Y = y) = \frac{\theta^y(1-\theta)^{n-y}}{\binom{n}{y}\theta^y(1-\theta)^{n-y}} = \binom{n}{y}^{-1} \quad (\textstyle\sum x_i = y),$$

free of $\theta$: $\sum X_i$ is sufficient.

**Theorem 8.2 (Factorization)**

$Y = u(\mathbf{X})$ is sufficient for $\theta \in \Omega$ if and only if there are functions $k_1, k_2$ with

$$\prod_{i=1}^{n} f(x_i;\theta) = k_1(u(\mathbf{x}),\theta)\, k_2(\mathbf{x}) \qquad \text{for all } \mathbf{x} \text{ and } \theta \in \Omega.$$

**Proof.** Discrete case.

($\Leftarrow$) For $\mathbf{x}$ with $u(\mathbf{x}) = y$,

$$P_\theta(\mathbf{X}=\mathbf{x} \mid Y=y) = \frac{P_\theta(\mathbf{X}=\mathbf{x})}{\sum_{\mathbf{z}:u(\mathbf{z})=y} P_\theta(\mathbf{X}=\mathbf{z})} = \frac{k_1(y,\theta)k_2(\mathbf{x})}{k_1(y,\theta)\sum_{\mathbf{z}:u(\mathbf{z})=y} k_2(\mathbf{z})} = \frac{k_2(\mathbf{x})}{\sum_{u(\mathbf{z})=y} k_2(\mathbf{z})},$$

free of $\theta$ (0 if $u(\mathbf{x}) \neq y$).

($\Rightarrow$) $k_2(\mathbf{x}) := P_\theta(\mathbf{X}=\mathbf{x} \mid Y=u(\mathbf{x}))$ is free of $\theta$; with $k_1(y,\theta) = P_\theta(Y=y)$,

$$P_\theta(\mathbf{X}=\mathbf{x}) = P_\theta(\mathbf{X}=\mathbf{x}, Y=u(\mathbf{x})) = k_2(\mathbf{x})\, k_1(u(\mathbf{x}),\theta).$$

■

**Example 8.6**

(a) Poisson($\theta$): $\prod f = e^{-n\theta}\theta^{\sum x_i} \cdot \prod(1/x_i!)$, so $\sum X_i$ is sufficient.

(b) Exp($\theta$): $\prod f = \theta^{-n}e^{-\sum x_i/\theta} \cdot \prod \mathbf{1}\{x_i > 0\}$, so $\sum X_i$ is sufficient.

(c) Gamma($\alpha,\beta$): $\prod f = \{\Gamma(\alpha)\beta^\alpha\}^{-n}(\prod x_i)^{\alpha-1}e^{-\sum x_i/\beta} \cdot \prod \mathbf{1}\{x_i > 0\}$, so $(\sum X_i, \prod X_i)$ is sufficient for $(\alpha,\beta)$.

**Theorem 8.3 (Exponential families)**

If $f(x;\theta) = \exp\{\sum_{j=1}^{k} g_j(\theta)T_j(x) - A(g(\theta)) + S(x)\}$ for $x \in \mathcal{X}$, with support $\mathcal{X}$ not depending on $\theta$, then $(\sum_i T_1(X_i), \ldots, \sum_i T_k(X_i))$ is sufficient for $\theta$.

**Proof.** $\prod f(x_i;\theta) = \exp\{\sum_j g_j(\theta)\sum_i T_j(x_i) - nA(g(\theta))\} \cdot \exp\{\sum S(x_i)\}\prod \mathbf{1}_{\mathcal{X}}(x_i)$; apply the factorization theorem. ■

**Theorem 8.4**

If $Y$ is sufficient for $\theta$ and $W = g(Y)$ with $g$ one-to-one, then $W$ is sufficient.

**Proof.** $\{W=w\} = \{Y = g^{-1}(w)\}$, so $P_\theta(\mathbf{X} \in A \mid W=w) = P_\theta(\mathbf{X} \in A \mid Y = g^{-1}(w))$, which does not depend on $\theta$. ■

**Example 8.7**

(a) $N(\mu,\sigma^2)$, $T = (x, x^2)$: $(\sum X_i, \sum X_i^2)$ and its one-to-one image $(\bar{X}, S^2)$ are sufficient.
(b) Beta($\alpha$, 1): $f = \exp\{(\alpha-1)\log x + \log\alpha\}$ on $(0,1)$, so $\sum \log X_i$ is sufficient.
(c) Support depending on $\theta$: use indicators. Two-parameter exponential $\frac{1}{\sigma}e^{-(x-\mu)/\sigma}\mathbf{1}\{x \geq \mu\}$:

$$\prod f(x_i;\mu,\sigma) = \sigma^{-n}e^{-(\sum x_i - n\mu)/\sigma}\mathbf{1}\{\min x_i \geq \mu\},$$

so $(\sum X_i, X_{(1)})$ is sufficient; for $U[\theta_1,\theta_2]$, $\prod f = (\theta_2-\theta_1)^{-n}\mathbf{1}\{\theta_1 \leq x_{(1)}, x_{(n)} \leq \theta_2\}$ gives $(X_{(1)}, X_{(n)})$.

**Theorem 8.5**

If the MLE $\hat{\theta}$ exists uniquely, it is a function of any sufficient statistic $S = u(\mathbf{X})$.

**Proof.** By factorization $\hat{\theta}(\mathbf{x}) = \arg\max_\theta k_1(u(\mathbf{x}), \theta)$ depends on $\mathbf{x}$ only through $u(\mathbf{x})$. ■

A sufficient unique MLE is then a function of every sufficient statistic: **minimal sufficient**, as in Examples 8.5–8.7.

**Theorem 8.6 (Rao–Blackwell)**

Let $Y$ be sufficient for $\theta$, $\hat{\eta}$ an estimator of $\eta(\theta)$ with finite variance, and $\hat{\eta}^{RB} = E[\hat{\eta} \mid Y]$. Then $\hat{\eta}^{RB}$ is a statistic, $E_\theta \hat{\eta}^{RB} = E_\theta \hat{\eta}$, and

$$MSE(\hat{\eta}^{RB}, \theta) \le MSE(\hat{\eta}, \theta) \qquad \text{for all } \theta.$$

**Proof.** The conditional law of $\mathbf{X}$ given $Y$ is free of $\theta$, so $\hat{\eta}^{RB}$ is a statistic; $E\hat{\eta}^{RB} = E\hat{\eta}$ and

$$\operatorname{Var} \hat{\eta} = E[\operatorname{Var}(\hat{\eta} \mid Y)] + \operatorname{Var}(E[\hat{\eta} \mid Y]) \ge \operatorname{Var} \hat{\eta}^{RB}.$$

With equal bias, $MSE = \operatorname{Var} + \text{bias}^2$ gives the claim. ■

**Example 8.8**

$U[0, \theta]$: $Y = X_{(n)}$ is sufficient; improve $2\bar{X}$ (MSE $\theta^2/(3n)$). Given $X_{(n)} = y$, the other order statistics are those of $n-1$ uniforms on $(0, y)$, so $E(X_{(r)} \mid X_{(n)} = y) = \frac{r}{n} y$, $r \le n-1$:

$$\hat{\theta}^{RB} = \frac{2}{n} E\Big( \sum_{r=1}^{n} X_{(r)} \,\Big|\, X_{(n)} \Big) = \frac{2}{n} \Big( \sum_{r=1}^{n-1} \frac{r}{n} + 1 \Big) X_{(n)} = \frac{n+1}{n} X_{(n)},$$

with $MSE = \int_0^\theta \left( \frac{n+1}{n} x - \theta \right)^2 \frac{n x^{n-1}}{\theta^n} dx = \frac{\theta^2}{n(n+2)} \le \frac{\theta^2}{3n}$.

## 8.3 Minimum variance unbiased estimation

**Motivation.** Conditioning on a complete sufficient statistic yields the unique best unbiased estimator.

**Definition (Completeness)**

$Y$ is **complete** for $\theta \in \Omega$ if $E_\theta g(Y) = 0$ for all $\theta$ implies $P_\theta(g(Y) = 0) = 1$ for all $\theta$; complete and sufficient: a **complete sufficient statistic**.

**Theorem 8.7 (Lehmann–Scheffé)**

Let $Y$ be complete sufficient for $\theta \in \Omega$.

(a) If $\hat{\eta}_0$ is any unbiased estimator of $\eta(\theta)$, then $E[\hat{\eta}_0 \mid Y]$ is the UMVUE of $\eta(\theta)$.

(b) If $\delta(Y)$ is unbiased for $\eta(\theta)$, then $\delta(Y)$ is the UMVUE.

**Proof.** (a) $\hat{\eta}_0^{RB} = E[\hat{\eta}_0 \mid Y]$ is unbiased. For any unbiased $\hat{\eta}_1$, $\operatorname{Var} \hat{\eta}_1^{RB} \le \operatorname{Var} \hat{\eta}_1$ (Rao–Blackwell); $\hat{\eta}_0^{RB} - \hat{\eta}_1^{RB}$ is a function of $Y$ with mean 0, so it is 0 a.s. by completeness, and $\operatorname{Var} \hat{\eta}_0^{RB} \le \operatorname{Var} \hat{\eta}_1$.

(b) Take $\hat{\eta}_0 = \delta(Y)$. ■

Recipe: find a complete sufficient $Y$, then an unbiased function of $Y$, or condition a simple unbiased estimator on $Y$.

**Example 8.9 (Bernoulli)**

$Y = \sum X_i \sim B(n, \theta)$. If $\sum_y g(y)\binom{n}{y}\theta^y(1-\theta)^{n-y} = 0$ for all $\theta \in (0,1)$, then with $\lambda = \theta/(1-\theta)$ the polynomial $\sum_y g(y)\binom{n}{y}\lambda^y$ vanishes on $(0,\infty)$, so $g \equiv 0$. So $Y$ is complete sufficient and $\bar{X}$ is the UMVUE.

**Example 8.10 (Poisson and exponential)**

(a) Poisson: $Y \sim \text{Poisson}(n\theta)$; $\sum_y g(y)\lambda^y/y! = 0$ for all $\lambda > 0$ forces $g \equiv 0$ (power series). $\bar{X}$ is the UMVUE.
(b) Exponential: $Y \sim \text{Gamma}(n, \theta)$; $\int_0^\infty g(y)y^{n-1}e^{-\lambda y}dy = 0$ for all $\lambda > 0$ forces $g(y)y^{n-1} = 0$ (Laplace transform). $\bar{X}$ is the UMVUE.

For exponential families (stated without proof, via Laplace transforms):

**Theorem 8.8**

Let $f(x;\eta) = \exp\{\sum_{j=1}^k \eta_j T_j(x) - A(\eta) + S(x)\}$ satisfy:
(i) the support does not depend on $\eta$;
(ii) the parameter space $N$ contains an open box $(a_1, b_1) \times \cdots \times (a_k, b_k)$;
(iii) $\mathbf{c}'\mathbf{T}(X)$ is not constant for $\mathbf{c} \neq \mathbf{0}$. Then $\left(\sum_i T_1(X_i), \ldots, \sum_i T_k(X_i)\right)$ is complete sufficient.

**Example 8.11**

(a) $N(\mu, \sigma^2)$: the natural parameter ranges over $\mathbb{R} \times (-\infty, 0)$, so $(\bar{X}, S^2)$ is complete sufficient, and unbiased: the UMVUEs.
(b) $N(\mu, \mu^2)$, $\mu > 0$: the natural parameter lies on a curve (no open box). Indeed $E_\mu Y_2 = 2n\mu^2$ and $E_\mu Y_1^2 = n(n+1)\mu^2$, so $g = \frac{Y_2}{2n} - \frac{Y_1^2}{n(n+1)} \neq 0$ has mean 0: $(\sum X_i, \sum X_i^2)$ is not complete.
(c) $U[0,\theta]$: $Y = X_{(n)}$ has pdf $ny^{n-1}/\theta^n$; $\int_0^\theta g(y)y^{n-1}dy = 0$ for all $\theta$ gives $g(\theta)\theta^{n-1} = 0$. So $\frac{n+1}{n}X_{(n)}$ is the UMVUE (Example 8.8).

**Example 8.12 (Bernoulli variance)**

$\eta = \theta(1-\theta)$, $n \geq 2$: $X_1(1-X_2)$ is unbiased, and

$$E[X_1(1-X_2) \mid Y = y] = \frac{P(X_1 = 1, X_2 = 0, \sum_{i\geq 3} X_i = y-1)}{P(Y = y)} = \frac{\binom{n-2}{y-1}}{\binom{n}{y}} = \frac{y(n-y)}{n(n-1)}.$$

so the UMVUE is $\frac{n}{n-1}\hat{\theta}(1-\hat{\theta})$, $\hat{\theta} = Y/n$.

**Example 8.13 (A function of a Poisson mean)**

$\eta = e^{-2\theta}$, Poisson($\theta$): find $\delta$ with $E_\theta \delta(Y) = e^{-2\theta}$, $Y \sim$ Poisson($n\theta$):

$$\sum_{y\geq 0} \delta(y) \frac{e^{-n\theta}(n\theta)^y}{y!} = e^{-2\theta} \iff \sum_y \frac{\delta(y)}{y!}\lambda^y = e^{(1-2/n)\lambda} = \sum_y \frac{(1-2/n)^y}{y!}\lambda^y,$$

$\lambda = n\theta$; comparing coefficients, $\delta(y) = (1-2/n)^y \approx e^{-2\bar{X}}$ (the MLE) for large $n$. For $n = 1$ the UMVUE $(-1)^{X_1}$ takes the value $-1$ for an estimand in $(0, 1)$: unbiasedness can be absurd.

**Example 8.14 (Reliability for a normal mean)**

$N(\theta, 1)$, $n \geq 2$, $\eta = P_\theta(X_1 > a)$: $\mathbf{1}\{X_1 > a\}$ is unbiased and $\bar{X}$ complete sufficient, so the UMVUE is $P(X_1 > a \mid \bar{X})$. As $X_1 - \bar{X} \sim N(0, \frac{n-1}{n})$ is independent of $\bar{X}$ (Theorem 4.5 in Volume I),

$$P(X_1 > a \mid \bar{X} = y) = P(X_1 - \bar{X} > a - y) = 1 - \Phi\Big(\sqrt{\tfrac{n}{n-1}}\,(a-y)\Big),$$

giving $1 - \Phi(\sqrt{n/(n-1)}\,(a - \bar{X}))$.

A general source of such independence:

**Theorem 8.9 (Basu)**

If $Y$ is complete sufficient for $\theta$, and the distribution of a statistic $Z$ does not depend on $\theta$ ($Z$ is **ancillary**), then $Y$ and $Z$ are independent.

**Proof.** Fix $B$. $P(Z \in B \mid Y)$ is free of $\theta$ (sufficiency) and $p_B = P_\theta(Z \in B)$ too (ancillarity); $E_\theta[P(Z \in B \mid Y) - p_B] = 0$ for all $\theta$, so completeness gives $P(Z \in B \mid Y) = p_B$ a.s. ■

**Example 8.15 (Reliability for an exponential mean)**

Exp($\theta$), $\eta = e^{-a/\theta}$: the UMVUE is $P(X_1 > a \mid Y)$, $Y = \sum X_i$. With $X_i = \theta Z_i$, $Z_i \overset{\text{i.i.d.}}{\sim}$ Exp(1), $X_1/Y \sim$ Beta$(1, n-1)$ is ancillary, hence independent of $Y$ (Basu):

$$P(X_1 > a \mid Y = y) = P\Big(\frac{X_1}{Y} > \frac{a}{y}\Big) = \int_{a/y}^1 (n-1)(1-z)^{n-2}dz = \Big(1 - \frac{a}{y}\Big)^{n-1}\mathbf{1}\{y > a\},$$

so the UMVUE is $(1 - \frac{a}{n\bar{X}})^{n-1}\mathbf{1}\{n\bar{X} > a\}$, close to the MLE $e^{-a/\bar{X}}$ for large $n$.

## 8.4 Asymptotic comparison

Typically squared bias is $O(n^{-2})$ and variance of order $n^{-1}$, so for large $n$ compare limiting variances.

**Definition (Asymptotic relative efficiency)**

If $\sqrt{n}(\hat{\theta}^i_n - \theta) \xrightarrow{d} N(0, \sigma_i^2(\theta))$, $i = 1, 2$, the **asymptotic relative efficiency** of $\hat{\theta}^1$ with respect to $\hat{\theta}^2$ is

$$ARE(\{\hat{\theta}^1_n\}, \{\hat{\theta}^2_n\}) = \frac{\sigma_2^2(\theta)}{\sigma_1^2(\theta)}.$$

Above 1, $\hat\theta^1$ is more precise: for a 95% error bound $1.96\sigma_i/\sqrt{n_i} = d$, $n_i \approx (1.96/d)^2\sigma_i^2$, so $n_2/n_1 = ARE$.

**Example 8.16 (Location families)**

Location model $f(x-\theta)$, $f$ symmetric about 0. By Chapter 5 and Theorem 6.13,

$$\sqrt{n}(\hat\theta^{\text{med}} - \theta) \xrightarrow{d} N\Big(0, \frac{1}{4f(0)^2}\Big), \qquad \sqrt{n}(\bar X - \theta) \xrightarrow{d} N(0, \sigma_f^2), \qquad \sqrt{n}(\hat\theta^{\text{MLE}} - \theta) \xrightarrow{d} N(0, I_f^{-1}),$$

with $\sigma_f^2 = \int x^2 f$, $I_f = \int (f'/f)^2 f$:

| | $N(\theta,1)$ | $L(\theta,1)$ | $DE(\theta,1)$ |
|---|---|---|---|
| $1/(4f(0)^2)$ (median) | $\pi/2$ | 4 | 1 |
| $\sigma_f^2$ (mean) | 1 | $\pi^2/3$ | 2 |
| $I_f^{-1}$ (MLE) | 1 | 3 | 1 |
| $ARE$(mean, median) | $\pi/2$ | $12/\pi^2$ | 1/2 |
| $ARE$(median, MLE) | $2/\pi$ | 3/4 | 1 |

For logistic data: MLE > mean > median; the median needs about 20% more observations than the mean.

**Example 8.17 (Beta$(\alpha,1)$)**

$m_1 = \alpha/(\alpha+1)$, so $\hat\alpha^{\text{MME}} = \bar X/(1-\bar X)$; with $\sigma^2 = \frac{\alpha}{\alpha+2} - \frac{\alpha^2}{(\alpha+1)^2}$ and $\dot g(m_1) = (\alpha+1)^2$ for $g(m) = m/(1-m)$, the delta method gives

$$\sqrt{n}(\hat\alpha^{\text{MME}} - \alpha) \xrightarrow{d} N\Big(0, \frac{\alpha(\alpha+1)^2}{\alpha+2}\Big).$$

With $T = \log x$, the MLE solves $\sum \log X_i + n/\alpha = 0$: $\hat\alpha^{\text{MLE}} = -1/\overline{\log X}$, $I(\alpha) = 1/\alpha^2$, limit $N(0,\alpha^2)$. Hence

$$ARE(\{\hat\alpha^{\text{MME}}\}, \{\hat\alpha^{\text{MLE}}\}) = \frac{\alpha^2(\alpha+2)}{\alpha(\alpha+1)^2} = \frac{\alpha(\alpha+2)}{(\alpha+1)^2} < 1.$$

Why the MLE wins:

**Theorem 8.10 (Information inequality)**

Assume the regularity conditions (R0)–(R5) of Chapter 6. For any estimator $\hat\eta_n = \hat\eta_n(X_1,\dots,X_n)$ of a real quantity with finite variance,

$$\operatorname{Var}_\theta(\hat\eta_n) \ge \Big(\frac{\partial}{\partial\theta} E_\theta \hat\eta_n\Big)' [nI(\theta)]^{-1} \Big(\frac{\partial}{\partial\theta} E_\theta \hat\eta_n\Big).$$

**Proof.** One-dimensional $\theta$: by Cauchy–Schwarz,

$$\operatorname{Var}_\theta(\hat\eta_n)\operatorname{Var}_\theta(\dot l_n(\theta)) \ge \{\operatorname{Cov}_\theta(\hat\eta_n, \dot l_n(\theta))\}^2.$$

By Theorem 6.12, $E_\theta \dot l_n(\theta) = 0$ and $\operatorname{Var}_\theta \dot l_n(\theta) = nI(\theta)$; by (R4)

$$\operatorname{Cov}_\theta(\hat\eta_n, \dot l_n(\theta)) = E_\theta[\hat\eta_n \dot l_n(\theta)] = \int \hat\eta_n(\mathbf{x}) \frac{\partial}{\partial\theta} \prod f(x_i;\theta)\, d\mathbf{x} = \frac{\partial}{\partial\theta} E_\theta \hat\eta_n.$$

Substitute. For vector $\theta$, with $\Sigma_{11} = \operatorname{Var}\hat{\eta}_n$, $\Sigma_{12} = \left(\partial_\theta E_\theta \hat{\eta}_n\right)'$, $\Sigma_{22} = nI(\theta)$,

$$\Sigma_{11} - \Sigma_{12}\Sigma_{22}^{-1}\Sigma_{21} = \operatorname{Var}\left(\hat{\eta}_n - \Sigma_{12}\Sigma_{22}^{-1}\dot{l}_n(\theta)\right) \geq 0.$$

■

**Corollary 8.11 (Cramér–Rao bound)**

If $\hat{\theta}^{UE}$ is unbiased for $\theta \in \mathbb{R}^k$, then $\mathbf{c}' \operatorname{Var}_\theta(\hat{\theta}^{UE})\mathbf{c} \geq \mathbf{c}' I(\theta)^{-1}\mathbf{c}/n$ for all $\mathbf{c}$; i.e. $\operatorname{Var}_\theta(\hat{\theta}^{UE}) - I(\theta)^{-1}/n$ is nonnegative definite. For $k = 1$, $\operatorname{Var}_\theta(\hat{\theta}^{UE}) \geq 1/\{nI(\theta)\}$.

**Proof.** Apply Theorem 8.10 to $\hat{\eta} = \mathbf{c}'\hat{\theta}^{UE}$, for which $\partial_\theta E_\theta \hat{\eta} = \mathbf{c}$. ■

**Example 8.18**

Beta$(\alpha, 1)$: $Y = \sum \log X_i$ is complete sufficient and $-\alpha Y \sim$ Gamma$(n, 1)$, so

$$E\left[\frac{1}{-Y}\right] = \frac{\alpha}{n-1}, \qquad \operatorname{Var}\left[\frac{1}{-Y}\right] = \frac{\alpha^2}{(n-1)^2(n-2)}.$$

So $\hat{\alpha}^{\text{UMVUE}} = (n-1)/(-Y) = \frac{n-1}{n}\hat{\alpha}^{\text{MLE}}$ has variance $\alpha^2/(n-2)$, above the bound $\alpha^2/n$ but asymptotically equal, with the MLE's limit $N(0, \alpha^2)$.

So $\operatorname{Var}\{\sqrt{n}(\hat{\theta}^{UE} - \theta)\} \geq 1/I(\theta)$, suggesting every asymptotically normal estimator has limiting variance at least $1/I(\theta)$ (**Fisher's conjecture**); false in general, true under extra conditions such as locally uniform convergence. In this sense the MLE is **asymptotically efficient**.

**Example 8.19 (Gamma)**

Gamma$(\alpha, \beta)$: the MME (Example 6.1(d)) satisfies

$$\sqrt{n}(\hat{\theta}^{\text{MME}} - \theta) \xrightarrow{d} N(\mathbf{0}, \Sigma(\theta)), \qquad \Sigma(\theta) = \begin{pmatrix} 2\alpha(\alpha+1) & -2(\alpha+1)\beta \\ -2(\alpha+1)\beta & (2+3/\alpha)\beta^2 \end{pmatrix},$$

and the MLE (or one-step, Example 6.16) has limit covariance $I(\theta)^{-1}$:

$$ARE(\hat{\alpha}^{\text{MME}}, \hat{\alpha}^{\text{MLE}}) = \frac{1}{2(\alpha+1)\{\alpha\dot{\Psi}(\alpha) - 1\}},$$

$$ARE(\hat{\beta}^{\text{MME}}, \hat{\beta}^{\text{MLE}}) = \frac{\alpha\dot{\Psi}(\alpha)}{(2\alpha+3)\{\alpha\dot{\Psi}(\alpha) - 1\}}.$$

About 0.39 and 0.51 at $\alpha = 1$, increasing to 1 as $\alpha \to \infty$ but always below 1.

# Chapter 9. Comparing Tests

## 9.1 Criteria for comparing tests

Simplest case: **simple** hypotheses, $\Omega = \{\theta_0, \theta_1\}$,

$$H_0 : \theta = \theta_0 \quad \text{vs} \quad H_1 : \theta = \theta_1.$$

A test $\phi$ has errors $E_{\theta_0}\phi(\mathbf{X})$ (type I) and $E_{\theta_1}[1-\phi(\mathbf{X})]$ (type II); reducing one increases the other, so combine them:

- **maximum error**: $\max\{E_{\theta_0}\phi, E_{\theta_1}(1-\phi)\}$;
- **Bayes error**: $\pi_0 E_{\theta_0}\phi + \pi_1 E_{\theta_1}(1-\phi)$, priors $\pi_0, \pi_1 > 0$, $\pi_0 + \pi_1 = 1$.

Let $p_\theta(\mathbf{x}) = \prod f(x_i;\theta)$.

**Theorem 9.1 (Bayes test for simple hypotheses)**

The test

$$\phi^\pi(\mathbf{x}) = \begin{cases} 1, & p_{\theta_1}(\mathbf{x})/p_{\theta_0}(\mathbf{x}) > \pi_0/\pi_1, \\ 0, & p_{\theta_1}(\mathbf{x})/p_{\theta_0}(\mathbf{x}) < \pi_0/\pi_1 \end{cases}$$

(arbitrary on equality; positive/0 $= +\infty$) minimizes the Bayes error.

**Proof.** The Bayes error is

$$\pi_0 \int \phi p_{\theta_0} + \pi_1\Big(1 - \int \phi p_{\theta_1}\Big) = \pi_1 - \int \phi(\mathbf{x})\{\pi_1 p_{\theta_1}(\mathbf{x}) - \pi_0 p_{\theta_0}(\mathbf{x})\}\, d\mathbf{x}.$$

minimized by $\phi = 1$ where the brace is positive and 0 where negative. ■

**Example 9.1**

$N(\mu, 1)$, $H_0 : \mu = \mu_0$ vs $H_1 : \mu = \mu_1$:

$$\frac{p_{\mu_1}(\mathbf{x})}{p_{\mu_0}(\mathbf{x})} = \exp\Big\{n(\mu_1 - \mu_0)\Big(\bar{x} - \frac{\mu_0 + \mu_1}{2}\Big)\Big\}.$$

For $\mu_1 > \mu_0$, reject when $\bar{x} > \frac{\mu_0+\mu_1}{2} + \frac{\log(\pi_0/\pi_1)}{n(\mu_1-\mu_0)}$ (symmetrically for $\mu_1 < \mu_0$); equal priors give the midpoint.

These criteria are symmetric in $H_0, H_1$; Chapter 7 instead bounds the type I error by $\alpha$ and maximizes the power $\gamma_\phi(\theta_1) = E_{\theta_1}\phi$.

**Definition (Most powerful test)**

For simple $H_0$ and $H_1$, a test $\phi^{MP}$ is a **most powerful (MP) test** of level $\alpha$ if (i) $E_{\theta_0}\phi^{MP} \le \alpha$, and (ii) $E_{\theta_1}\phi^{MP} \ge E_{\theta_1}\phi$ for every $\phi$ with $E_{\theta_0}\phi \le \alpha$.

**Theorem 9.2 (Neyman–Pearson lemma)**

A test of the form

$$\phi^*(\mathbf{x}) = \begin{cases} 1, & p_{\theta_1}(\mathbf{x})/p_{\theta_0}(\mathbf{x}) > k, \\ \gamma, & p_{\theta_1}(\mathbf{x})/p_{\theta_0}(\mathbf{x}) = k, \\ 0, & p_{\theta_1}(\mathbf{x})/p_{\theta_0}(\mathbf{x}) < k, \end{cases} \qquad k \geq 0,\ 0 \leq \gamma \leq 1,$$

with size $E_{\theta_0}\phi^* = \alpha$ is a most powerful level-$\alpha$ test.

**Proof.** For any $\phi$, $E_{\theta_1}\phi - kE_{\theta_0}\phi = \int \phi\,(p_{\theta_1} - kp_{\theta_0})\,d\mathbf{x}$, so

$$E_{\theta_1}\phi - kE_{\theta_0}\phi = \int \phi\,(p_{\theta_1} - kp_{\theta_0})\,d\mathbf{x}.$$

$$\{E_{\theta_1}\phi^* - kE_{\theta_0}\phi^*\} - \{E_{\theta_1}\phi - kE_{\theta_0}\phi\} = \int (\phi^* - \phi)(p_{\theta_1} - kp_{\theta_0})\,d\mathbf{x} \geq 0,$$

since $\phi^* = 1 \geq \phi$ where $p_{\theta_1} > kp_{\theta_0}$ and $\phi^* = 0 \leq \phi$ where $p_{\theta_1} < kp_{\theta_0}$. So if $E_{\theta_0}\phi \leq \alpha$,

$$E_{\theta_1}\phi^* - E_{\theta_1}\phi \geq k\{E_{\theta_0}\phi^* - E_{\theta_0}\phi\} = k\{\alpha - E_{\theta_0}\phi\} \geq 0.$$

■

Such $k, \gamma$ always exist, and every MP test has this form a.s.; $\gamma$ may depend on $\mathbf{x}$.

**Example 9.2**

Poisson($\theta$), $n = 50$, $H_0 : \theta = 0.1$ vs $H_1 : \theta = 0.05$: the ratio $e^{2.5}(1/2)^{\sum x_i}$ decreases in $S = \sum x_i$, so reject for $S \leq c - 1$, randomize at $S = c$, with $E_{0.1}\phi^* = 0.05$: $c = 2$, $\gamma = 0.114$ (Example 7.3).

## 9.2 Uniformly most powerful tests

**Motivation.** With monotone likelihood ratio, the Neyman–Pearson test is best against all one-sided alternatives at once.

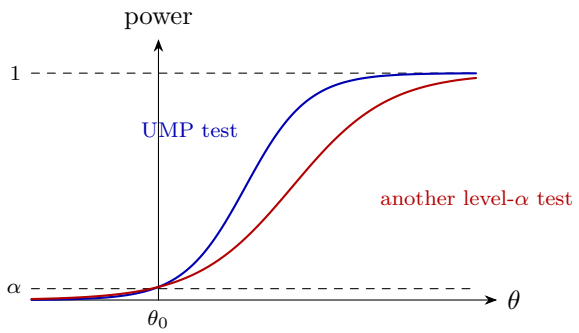



**Figure.** *Power functions for $H_0 : \theta \leq \theta_0$ vs $H_1 : \theta > \theta_0$: both size $\alpha$; the UMP test has larger power at every $\theta > \theta_0$.*

For composite hypotheses we want large power at every $\theta_1 \in \Omega_1$.

**Definition (UMP test)**

$\phi^{UMP}$ is a **uniformly most powerful (UMP) test** of level $\alpha$ if (i) $\max_{\theta\in\Omega_0} E_\theta\phi^{UMP} \le \alpha$, and (ii) $E_{\theta_1}\phi^{UMP} \ge E_{\theta_1}\phi$ for all $\theta_1 \in \Omega_1$ and every $\phi$ with $\max_{\Omega_0} E_\theta\phi \le \alpha$.

**Example 9.3**

$N(\mu,1)$, $H_0:\mu=\mu_0$ vs $H_1:\mu>\mu_0$. For each $\mu_1>\mu_0$ the ratio increases in $\bar x$ (Example 9.1), so the MP test is $\bar x \ge \mu_0 + z_\alpha/\sqrt{n}$, free of $\mu_1$: UMP,

$$\phi^*(\mathbf{x}) = \mathbf{1}\{\bar x - \mu_0 \ge z_\alpha/\sqrt{n}\}.$$

**Example 9.4**

It is also UMP for $H_0^-:\mu\le\mu_0$: its power $P(Z\ge\sqrt{n}(\mu_0-\mu)+z_\alpha)$ increases in $\mu$, so it has level $\alpha$ for $H_0^-$, and every level-$\alpha$ test for $H_0^-$ has level $\alpha$ at $\mu_0$.

**Theorem 9.3 (UMP one-sided tests in exponential families)**

Let $f(x;\theta)=\exp\{g(\theta)T(x)-B(\theta)+S(x)\}$, $x\in\mathcal{X}$, $\theta\in\Omega\subseteq\mathbb{R}$, with $g$ increasing. For $H_0:\theta\le\theta_0$ vs $H_1:\theta>\theta_0$, the test

$$\phi^*(\mathbf{x}) = \begin{cases} 1, & \sum T(x_i) > c, \\ \gamma, & \sum T(x_i) = c, \\ 0, & \sum T(x_i) < c, \end{cases} \qquad E_{\theta_0}\phi^* = \alpha,$$

is UMP of level $\alpha$. (For $H_0:\theta\ge\theta_0$ vs $H_1:\theta<\theta_0$, reverse the inequalities.)

**Proof.**

*Step 1.* For $\theta_1>\theta_0$,

$$\frac{p_{\theta_1}(\mathbf{x})}{p_{\theta_0}(\mathbf{x})} = \exp\Big\{(g(\theta_1)-g(\theta_0))\sum T(x_i) - n(B(\theta_1)-B(\theta_0))\Big\}$$

increases in $\sum T(x_i)$: $\phi^*$ is the Neyman–Pearson test, free of $\theta_1$, so UMP for $H_0:\theta=\theta_0$.

*Step 2.* $E_\theta\phi^*$ is nondecreasing: for $\theta'<\theta''$ and $\alpha'=E_{\theta'}\phi^*$, $\phi^*$ is MP at level $\alpha'$ for $\theta'$ vs $\theta''$, and beats the trivial $\phi\equiv\alpha'$: $E_{\theta''}\phi^*\ge\alpha'$.

*Step 3.* So $\max_{\theta\le\theta_0} E_\theta\phi^*=\alpha$, and any level-$\alpha$ test of $H_0:\theta\le\theta_0$ has level $\alpha$ at $\theta_0$; apply Step 1. ■

**Example 9.5**

(a) Poisson: $g=\log$ increasing; for $H_0:\theta\ge0.1$ vs $H_1:\theta<0.1$, $n=50$, the UMP level-0.05 test is that of Example 9.2.
(b) Exponential: $g(\theta)=-1/\theta$ increasing; for $H_0:\theta\le\theta_0$ reject when $\bar x/\theta_0 \ge \chi^2_\alpha(2n)/(2n)$ (as $2\sum X_i/\theta_0 \sim \chi^2(2n)$).

**Example 9.6 (No UMP two-sided test)**

$N(\mu,1)$, $H_0:\mu=\mu_0$ vs $H_1:\mu\neq\mu_0$. A UMP test would be MP against both $\mu_1<\mu_0$ and $\mu_2>\mu_0$, hence (converse of Neyman–Pearson) both $\mathbf{1}\{\bar{x}\le c_1\}$ and $\mathbf{1}\{\bar{x}\ge c_2\}$ of size $\alpha$: impossible.

**Example 9.7 (Optimality of the two-sided test)**

The LRT $\phi^*=\mathbf{1}\{\sqrt{n}|\bar{x}-\mu_0|\ge z_{\alpha/2}\}$ has power $1-\Phi(z_{\alpha/2}-\delta)+\Phi(-z_{\alpha/2}-\delta)$, $\delta=\sqrt{n}(\mu-\mu_0)$, so (a) $E_{\mu_0}\phi^*=\alpha$ and $\frac{d}{d\mu}E_\mu\phi^*\big|_{\mu_0}=0$. Claim (b): among tests with these two properties, $\phi^*$ is most powerful at every $\mu_1\neq\mu_0$.
With $\dot{p}_{\mu_0}=\partial_\mu p_\mu|_{\mu_0}$, $\frac{d}{d\mu}E_\mu\phi|_{\mu_0}=E_{\mu_0}[\phi\,\dot{p}_{\mu_0}/p_{\mu_0}]$, and for constants $k_1,k_2$

$$E_{\mu_1}\phi-k_1E_{\mu_0}\phi-k_2E_{\mu_0}\Big[\phi\frac{\dot{p}_{\mu_0}}{p_{\mu_0}}\Big]=\int\phi\,\{p_{\mu_1}-k_1p_{\mu_0}-k_2\dot{p}_{\mu_0}\}\,d\mathbf{x}$$

is maximized by $\phi^{**}=\mathbf{1}\{p_{\mu_1}\ge k_1p_{\mu_0}+k_2\dot{p}_{\mu_0}\}$. As $\dot{p}_{\mu_0}/p_{\mu_0}=n(\bar{x}-\mu_0)$, this is $\{e^{a(\bar{x}-\mu_0)}\ge b_1+b_2(\bar{x}-\mu_0)\}$, the complement of an interval (convex exponential above a line). The side conditions make it symmetric: $\phi^{**}=\phi^*$. As in Neyman–Pearson, $E_{\mu_1}\phi^*\ge E_{\mu_1}\phi$ for every $\phi$ with the side conditions, for all $\mu_1$.
A test with $\max_{\Omega_0}E_\theta\phi\le\alpha\le\min_{\Omega_1}E_\theta\phi$ is **unbiased**; for smooth power functions this implies the side conditions, so the two-sided test is **UMP unbiased**.

## 9.3 Nonparametric tests and asymptotic comparison

Now let $X_1,\dots,X_n$ come from a continuous location model $f(x-\theta)$, $f$ symmetric about 0, strictly increasing cdf $F$; $\theta$ is the median.

**Example 9.8 (Sign test)**

For $H_0:\theta=\theta_0$ vs $H_1:\theta>\theta_0$, $S_n=\sum_i\mathbf{1}\{X_i>\theta_0\}\sim B(n,p(\theta))$, $p(\theta)=1-F(\theta_0-\theta)$, and $H_0$ is $p=\frac{1}{2}$. The test

$$\phi_S=\begin{cases}1, & S_n\ge c+1,\\ \gamma, & S_n=c,\\ 0, & S_n\le c-1,\end{cases}\qquad \sum_{k=c+1}^{n}\binom{n}{k}2^{-n}+\gamma\binom{n}{c}2^{-n}=\alpha,$$

has size $\alpha$ for every $f$. With $X_i=Z_i+\theta$, $E_\theta\phi_S$ increases in $\theta$ ($\phi_S$ is nondecreasing in each $x_i$), so $\phi_S$ also has level $\alpha$ for $H_0:\theta\le\theta_0$: the **sign test**.

A test whose size does not depend on the population form is **nonparametric** (distribution-free); its cost in efficiency is measured by the sample size for given power.

**Example 9.9 (Power of the sign test)**

By the normal approximation the sign test rejects when $S_n\ge np(\theta_0)+\sqrt{n}\sigma(\theta_0)z_\alpha$, $p(\theta_0)=\frac{1}{2}$, $\sigma^2(\theta)=p(\theta)(1-p(\theta))$; at $\theta_1>\theta_0$

$$\gamma_n(\theta_1)\approx 1-\Phi\Big(-\sqrt{n}\frac{p(\theta_1)-p(\theta_0)}{\sigma(\theta_1)}+\frac{\sigma(\theta_0)}{\sigma(\theta_1)}z_\alpha\Big)\to 1.$$

For $\theta_{1n} = \theta_0 + K/\sqrt{n}$, $p(\theta_{1n}) - p(\theta_0) \approx (\theta_{1n} - \theta_0)f(0)$, and

$$\gamma_n(\theta_{1n}) \approx 1 - \Phi\big(-2\sqrt{n}(\theta_{1n} - \theta_0)f(0) + z_\alpha\big).$$

so power $\gamma$ needs $n \approx (2f(0))^{-2}\big(\frac{z_\alpha + z_{1-\gamma}}{\theta_{1n} - \theta_0}\big)^2$.

**Theorem 9.4 (Approximate power and sample size)**

For $H_0 : \theta = \theta_0$ vs $H_1 : \theta > \theta_0$, let a test reject when $\sqrt{n}(T_n - \mu(\theta_0))/\sigma(\theta_0) \geq t_n$, with size $\alpha$, where

$$\sqrt{n}\,(T_n - \mu(\theta))/\sigma(\theta) \xrightarrow{d} N(0,1)$$

(uniformly for $\theta$ near $\theta_0$) and $\mu, \sigma$ are continuously differentiable. Then:

(a) for $\theta_{1n} = \theta_0 + K/\sqrt{n}$, $\gamma_n(\theta_{1n}) \approx 1 - \Phi\big(-\sqrt{n}(\theta_{1n} - \theta_0)\dot{\mu}(\theta_0)/\sigma(\theta_0) + z_\alpha\big)$;

(b) the sample size needed for power $\gamma$ at $\theta_{1n}$ is

$$N(T; \gamma, \theta_{1n}) \approx \Big(\frac{\dot{\mu}(\theta_0)}{\sigma(\theta_0)}\Big)^{-2}\Big(\frac{z_\alpha + z_{1-\gamma}}{\theta_{1n} - \theta_0}\Big)^2.$$

**Proof.** Size $\alpha$ and asymptotic normality at $\theta_0$ give $t_n \approx z_\alpha$. Under $\theta_1$,

$$P_{\theta_1}\Big(\sqrt{n}\frac{T_n - \mu(\theta_1)}{\sigma(\theta_1)} \geq -\sqrt{n}\frac{\mu(\theta_1) - \mu(\theta_0)}{\sigma(\theta_1)} + \frac{\sigma(\theta_0)}{\sigma(\theta_1)}t_n\Big) \approx 1 - \Phi\Big(-\sqrt{n}\frac{\mu(\theta_1) - \mu(\theta_0)}{\sigma(\theta_1)} + \frac{\sigma(\theta_0)}{\sigma(\theta_1)}z_\alpha\Big).$$

For $\theta_{1n}$, $\mu(\theta_{1n}) - \mu(\theta_0) \approx (\theta_{1n} - \theta_0)\dot{\mu}(\theta_0)$ and $\sigma(\theta_{1n}) \to \sigma(\theta_0)$: (a). For (b) solve $\sqrt{N}(\theta_{1n} - \theta_0)\dot{\mu}(\theta_0)/\sigma(\theta_0) - z_\alpha = z_{1-\gamma}$. ■

**Definition (Asymptotic relative efficiency of tests)**

For tests based on $T_{1n}, T_{2n}$ satisfying Theorem 9.4,

$$ARE(\{T_{1n}\}, \{T_{2n}\}) = \lim \frac{N(T_2; \gamma, \theta_{1n})}{N(T_1; \gamma, \theta_{1n})} = \frac{(\dot{\mu}_1(\theta_0)/\sigma_1(\theta_0))^2}{(\dot{\mu}_2(\theta_0)/\sigma_2(\theta_0))^2}.$$

**Example 9.10 (Sign test vs $t$ test)**

For $\frac{1}{\sigma}f(\frac{x-\mu}{\sigma})$, the sign test needs $N(S; \gamma, \mu_{1n}) \approx (2f(0)/\sigma)^{-2}\big(\frac{z_\alpha + z_{1-\gamma}}{\mu_{1n} - \mu_0}\big)^2$; the $t$ statistic has $\dot{\mu}/\sigma = 1/\sqrt{\operatorname{Var} X_1}$, $\operatorname{Var} X_1 = \sigma^2 \int z^2 f(z)dz$. So

$$ARE(\{S_n\}, \{T_n\}) = 4f(0)^2 \int z^2 f(z)\,dz,$$

as for median vs mean (Example 8.16):

| | $N(\mu, \sigma^2)$ | $L(\mu, \sigma)$ | $DE(\mu, \sigma)$ |
|---|---|---|---|
| $ARE(S, T)$ | $2/\pi = 0.637$ | $\pi^2/12 = 0.822$ | 2 |

Using ranks as well as signs gives a much more efficient nonparametric test.

**Definition (Signed rank statistic)**

With $R(|X_i-\theta_0|) = 1 + \sum_j \mathbf{1}\{|X_j-\theta_0| < |X_i-\theta_0|\}$, the **signed rank statistic** is

$$W_n = \sum_{i=1}^{n} \operatorname{sgn}(X_i-\theta_0)\, R(|X_i-\theta_0|),$$

and $W_n^+ = \sum_i \mathbf{1}\{X_i > \theta_0\} R(|X_i-\theta_0|)$ sums the ranks of positive deviations.

**Theorem 9.5 (Null distribution)**

Under $H_0 : \theta = \theta_0$:

(a) $W_n \overset{d}{=} \sum_{j=1}^{n} j\, S(j)$, with $S(1),\dots,S(n)$ i.i.d., $P(S(j)=1) = P(S(j)=-1) = \frac{1}{2}$;

(b) $EW_n = 0$, $\operatorname{Var} W_n = \sum j^2 = n(n+1)(2n+1)/6$, and $W_n/\sqrt{\operatorname{Var} W_n} \xrightarrow{d} N(0,1)$.

**Proof.** (a) Let $S(i) = \operatorname{sgn}(X_i-\theta_0)$, $R(i) = R(|X_i-\theta_0|)$. By symmetry

$$P(|X_1-\theta_0| \le x,\ \operatorname{sgn}(X_1-\theta_0) = \pm 1) = \tfrac{1}{2} P(|X_1-\theta_0| \le x),$$

so $|X_i-\theta_0| \perp \operatorname{sgn}(X_i-\theta_0)$; hence $S \perp R$, with i.i.d. fair signs. With inverse ranks $R^{-1}(j)$,

$$W_n = \sum_i S(i)R(i) = \sum_{j=1}^{n} S(R^{-1}(j))\, j.$$

For each permutation $\pi$, by $S \perp R$ and exchangeability of $S$,

$$P(S(R^{-1}(j)) = s_j\ \forall j,\ R = \pi) = P(S(\pi^{-1}(j)) = s_j\ \forall j) P(R=\pi) = P(S(j) = s_j\ \forall j) P(R=\pi).$$

Sum over $\pi$.

(b) Moments from (a). With $\sigma_n^2 = \sum j^2$, $Z_n = W_n/\sigma_n$ has cgf

$$\sum_{j=1}^{n} \log \frac{e^{jt/\sigma_n} + e^{-jt/\sigma_n}}{2} = \sum_j \log\Big(1 + \frac{j^2t^2}{2\sigma_n^2} + \dots\Big) = \frac{t^2}{2} + O\Big(\sum_j \frac{j^4}{\sigma_n^4}\Big),$$

and $\sum j^4/\sigma_n^4 = O(n^5/n^6) \to 0$. ∎

**Theorem 9.6**

(a) $W_n = 2W_n^+ - n(n+1)/2$.

(b) Under $H_0$, $W_n^+ \overset{d}{=} \sum_{j=1}^{n} jB_j$ with $B_j \overset{\text{i.i.d.}}{\sim} \text{Bernoulli}(\frac{1}{2})$, so $EW_n^+ = n(n+1)/4$, $\operatorname{Var} W_n^+ = n(n+1)(2n+1)/24$, and $(W_n^+ - n(n+1)/4)/\sqrt{n(n+1)(2n+1)/24} \xrightarrow{d} N(0,1)$.

**Proof.** (a) $\operatorname{sgn}(X_i-\theta_0) = 2\mathbf{1}\{X_i > \theta_0\} - 1$ (no ties), and $\sum R(i) = n(n+1)/2$.

(b) By (a) and Theorem 9.5(a), $W_n^+ \overset{d}{=} \sum j(S(j)+1)/2$. ∎

The null law of $W_n$ is free of $f$; large $W_n^+$ supports $\theta > \theta_0$. The **signed rank test** is

$$\phi_{SR} = \begin{cases} 1, & W_n^+ \ge c+1, \\ \gamma, & W_n^+ = c, \\ 0, & W_n^+ \le c-1, \end{cases} \qquad E_{\theta_0}\phi_{SR} = \alpha,$$

or for large $n$: reject when $(W_n^+ - n(n+1)/4)/\sqrt{n(n+1)(2n+1)/24} \ge z_\alpha$.

**Example 9.11**

$n = 10$, differences $1.8, -0.4, 2.5, 0.9, -1.1, 3.2, 0.6, 1.5, -0.2, 2.1$: absolute ranks $7, 2, 9, 4, 5, 10, 3, 6, 1, 8$, $W^+ = 47$. Null mean 27.5, SD 9.81, $z = 1.99 > 1.645$: reject at 5%. The sign test ($S = 7$, $P(S \geq 7) = 0.172$) does not.

**Theorem 9.7 (Walsh averages)**

(a) $W_n^+ = \sum_{1 \leq i \leq j \leq n} \mathbf{1}\{X_i + X_j > 2\theta_0\}$.

(b) The power function of $\phi_{SR}$ increases in $\theta$; hence $\max_{\theta \leq \theta_0} E_\theta \phi_{SR} = \alpha$, and $\phi_{SR}$ is a level-$\alpha$ test of $H_0 : \theta \leq \theta_0$ vs $H_1 : \theta > \theta_0$.

**Proof.** (a) With $Y_i = X_i - \theta_0$ sorted by absolute value, $|Y_{(1)}| < \cdots < |Y_{(n)}|$,

$$W_n^+ = \sum_i \mathbf{1}\{Y_{(i)} > 0\} \cdot i = \sum_i \mathbf{1}\{Y_{(i)} > 0\} + \sum_i \sum_{j<i} \mathbf{1}\{Y_{(i)} > 0\}.$$

For $j < i$, $Y_{(i)} > 0$ iff $Y_{(i)} + Y_{(j)} > 0$, so $W_n^+ = \sum_{i \leq j} \mathbf{1}\{Y_i + Y_j > 0\}$.

(b) By (a), $\phi_{SR}$ is nondecreasing in each $X_i$; with $X_i = Z_i + \theta$, $E_\theta \phi_{SR}$ is nondecreasing in $\theta$. ■

Behaviour under alternatives (stated without proof):

**Theorem 9.8**

Let $\mu(\theta) = \frac{1}{2} P_\theta(X_1 + X_2 > 2\theta_0)$ and $\sigma^2(\theta) = \mathrm{Cov}_\theta\left(\mathbf{1}\{X_1 + X_2 > 2\theta_0\}, \mathbf{1}\{X_1 + X_3 > 2\theta_0\}\right)$. Then $E_\theta W_n^+ \approx n^2 \mu(\theta)$, $\mathrm{Var}_\theta W_n^+ \approx n^3 \sigma^2(\theta)$, and $(W_n^+ - E_\theta W_n^+)/\sqrt{\mathrm{Var}_\theta W_n^+} \xrightarrow{d} N(0,1)$.

**Theorem 9.9 (Power of the signed rank test)**

For $\theta_{1n} = \theta_0 + K/\sqrt{n}$,

$$\gamma_{\phi_{SR}}(\theta_{1n}) \approx 1 - \Phi\Big( - \sqrt{n}(\theta_{1n} - \theta_0) \frac{\dot\mu(\theta_0)}{\sigma(\theta_0)} + z_\alpha \Big), \qquad \dot\mu(\theta_0) = \int f^2(x)\, dx, \quad \sigma^2(\theta_0) = \frac{1}{12},$$

and the required sample size is $N(W^+; \gamma, \theta_{1n}) \approx \left(\sqrt{12} \int f^2\right)^{-2} \left(\frac{z_\alpha + z_{1-\gamma}}{\theta_{1n} - \theta_0}\right)^2$.

**Proof.** Apply Theorem 9.4 to $T_n = W_n^+/n^2$. With $Z \sim F$,

$$\mu(\theta) = \frac{1}{2} P(Z_1 + Z_2 > 2(\theta_0 - \theta)) = \frac{1}{2} \int \{1 - F(2\theta_0 - 2\theta - x)\} f(x)\, dx,$$

so $\dot\mu(\theta_0) = \int f^2$. At $\theta_0$, $\mathbf{1}\{Z_1 + Z_2 > 0\}$ and $\mathbf{1}\{Z_1 + Z_3 > 0\}$ are conditionally independent given $Z_1$, each with mean $F(Z_1) =: U \sim U(0,1)$, so $\sigma^2(\theta_0) = \mathrm{Var}\, U = \frac{1}{12}$. ■

**Example 9.12 (Signed rank test vs $t$ test)**

Since $\int (\frac{1}{\sigma} f(\frac{x}{\sigma}))^2 dx = \frac{1}{\sigma} \int f^2$, Theorem 9.9 and Example 9.10 give

$$ARE(\{W_n^+\}, \{T_n\}) = 12 \Big( \int f^2(z)\, dz \Big)^2 \int z^2 f(z)\, dz.$$

| | $N(\mu, \sigma^2)$ | $L(\mu, \sigma)$ | $DE(\mu, \sigma)$ |
|---|---|---|---|
| $ARE(W^+, T)$ | $3/\pi = 0.955$ | $\pi^2/9 = 1.097$ | 1.5 |

Under normality the loss is below 5%; for heavier tails the signed rank test wins, with distribution-free size.

# Chapter 10. Analysis of Variance and Regression

## 10.1 One-way analysis of variance

The **one-way normal model** (Chapter 4) is

$$X_{ij} = \mu_i + e_{ij}, \qquad e_{ij} \overset{\text{i.i.d.}}{\sim} N(0, \sigma^2), \qquad i = 1, \dots, k,\ j = 1, \dots, n_i,$$

models a single **factor** with $k$ **levels** (**treatments**). With $n = \sum n_i$, overall effect $\bar\mu = \sum_i n_i \mu_i / n$ and treatment effects $\alpha_i = \mu_i - \bar\mu$,

$$X_{ij} = \bar\mu + \alpha_i + e_{ij}, \qquad \sum_i n_i \alpha_i = 0.$$

**Theorem 10.1 (UMVUEs)**

In the one-way normal model, the UMVUEs are

$$\hat\mu_i = \bar X_i = \frac{1}{n_i}\sum_j X_{ij}, \qquad \hat{\bar\mu} = \sum_i \frac{n_i}{n}\hat\mu_i, \qquad \hat\alpha_i = \hat\mu_i - \hat{\bar\mu}, \qquad \hat\sigma^2 = \frac{1}{n-k}\sum_{i,j}(X_{ij} - \bar X_i)^2.$$

**Proof.** The joint pdf $\exp\{-\frac{1}{2\sigma^2}\sum_{i,j}(x_{ij}-\mu_i)^2 - \frac{n}{2}\log(2\pi\sigma^2)\}$ is a $(k+1)$-parameter exponential family meeting Theorem 8.8, so $(\bar X_1, \dots, \bar X_k, \sum_{i,j}(X_{ij} - \bar X_i)^2)$ is complete sufficient; the estimators are unbiased functions of it (Theorem 4.9 in Volume I), hence UMVUEs (Lehmann–Scheffé). ■

The MLE of $\sigma^2$ is $\sum (X_{ij} - \bar X_i)^2 / n$; the other MLEs are the UMVUEs.

**Theorem 10.2 (Sampling distributions)**

(a) $\hat\mu_i = \bar X_i$ are independent, $\bar X_i \sim N(\mu_i, \sigma^2/n_i)$.
(b) $\hat\sigma^2$ is independent of $(\bar X_1, \dots, \bar X_k)$.
(c) $(n-k)\hat\sigma^2/\sigma^2 \sim \chi^2(n-k)$.

**Proof.** By Theorem 4.5 in Volume I per group, $\bar X_i \sim N(\mu_i, \sigma^2/n_i)$ is independent of $SS_i = \sum_j (X_{ij} - \bar X_i)^2$, $SS_i/\sigma^2 \sim \chi^2(n_i - 1)$; groups are independent and $(n-k)\hat\sigma^2 = \sum_i SS_i$. ■

### Simultaneous confidence intervals

Let $\hat\mu = (\hat\mu_1, \dots, \hat\mu_k)'$, $D = \mathrm{Var}(\hat\mu)/\sigma^2 = \mathrm{diag}(1/n_i)$, and for a rank-$r$ $k \times r$ matrix $C$, $\mathrm{col}(C) = \{C\mathbf{a} : \mathbf{a} \in \mathbb{R}^r\}$.

**Theorem 10.3 (Confidence sets and Scheffé intervals)**

(a) $P\{(C'\mu - C'\hat\mu)'(C'DC)^{-1}(C'\mu - C'\hat\mu) \le r\hat\sigma^2 F_\alpha(r, n-k)\} = 1 - \alpha$.

(b) $P\Big\{|\mathbf{c}'\mu - \mathbf{c}'\hat\mu| \le \sqrt{\textstyle\sum_i c_i^2 \hat\sigma^2/n_i}\sqrt{rF_\alpha(r,n-k)} \ \text{ for all } \mathbf{c} \in \mathrm{col}(C)\Big\} = 1 - \alpha$.

**Proof.** (a) $C'\hat\mu \sim N(C'\mu, \sigma^2 C'DC)$, so the quadratic form over $\sigma^2$ is $\chi^2(r)$ (Theorem 4.18 in Volume I), independent of $(n-k)\hat\sigma^2/\sigma^2 \sim \chi^2(n-k)$; the ratio is $F(r, n-k)$.

(b) For $\mathbf{c} = C\mathbf{a}$, $\mathbf{d} = C'(\mu - \hat\mu)$, $M = C'DC$, by Cauchy–Schwarz

$$\max_{\mathbf{c}\in\mathrm{col}(C),\mathbf{c}\neq\mathbf{0}} \frac{\{\mathbf{c}'(\mu-\hat\mu)\}^2}{\mathbf{c}'D\mathbf{c}} = \max_{\mathbf{a}\neq\mathbf{0}} \frac{(\mathbf{a}'\mathbf{d})^2}{\mathbf{a}'M\mathbf{a}} = \max_{\mathbf{b}\neq\mathbf{0}} \frac{(\mathbf{b}'M^{-1/2}\mathbf{d})^2}{\mathbf{b}'\mathbf{b}} = \mathbf{d}'M^{-1}\mathbf{d}.$$

so (b) and (a) are the same event. ■

$\sum c_i\mu_i$ with $\sum c_i = 0$ is a **contrast**, e.g. $\mu_i - \mu_l = \alpha_i - \alpha_l$; contrasts form a $(k-1)$-dimensional space:

**Theorem 10.4 (Simultaneous intervals for contrasts)**

(a) (Scheffé) With probability $1-\alpha$, for all $\mathbf{c}$ with $\sum c_i = 0$,

$$|\mathbf{c}'\alpha - \mathbf{c}'\hat{\alpha}| \le \sqrt{\textstyle\sum_i c_i^2\hat{\sigma}^2/n_i}\,\sqrt{(k-1)F_\alpha(k-1,n-k)}.$$

(b) In particular, with probability at least $1-\alpha$, for all $i \ne l$,

$$|(\alpha_i - \alpha_l) - (\hat{\alpha}_i - \hat{\alpha}_l)| \le \sqrt{(n_i^{-1}+n_l^{-1})\hat{\sigma}^2}\sqrt{(k-1)F_\alpha(k-1,n-k)}.$$

(c) (Bonferroni) With $m = k(k-1)/2$ and $\alpha^* = \alpha/m$, with probability at least $1-\alpha$, for all $i \ne l$,

$$|(\alpha_i - \alpha_l) - (\hat{\alpha}_i - \hat{\alpha}_l)| \le \sqrt{(n_i^{-1}+n_l^{-1})\hat{\sigma}^2}\; t_{\alpha^*/2}(n-k).$$

**Proof.** (a) Theorem 10.3(b) on the contrast space;

(b) is a sub-event.

(c) Per pair $r = 1$ and $F_{\alpha^*}(1,n-k) = t_{\alpha^*/2}(n-k)^2$, coverage $1-\alpha^*$; by Bonferroni $P(A_1\cap\cdots\cap A_m) \ge 1 - m\alpha^* = 1-\alpha$. ■

Bonferroni is shorter for few comparisons; Scheffé covers all contrasts. E.g. $n = 20$, $\alpha = 0.05$: $k = 3$ gives multipliers 2.680 (Scheffé) vs 2.654 (Bonferroni); $k = 7$, 4.182 vs 3.760.

### Testing equality of treatments

**Theorem 10.5 ($F$ test)**

With $SSB = \sum_i n_i(\bar{X}_i - \bar{X})^2$ (between) and $SSW = \sum_{i,j}(X_{ij} - \bar{X}_i)^2 = (n-k)\hat{\sigma}^2$ (within), the LRT of $H_0 : \mu_1 = \cdots = \mu_k$ rejects when

$$F_n = \frac{SSB/(k-1)}{SSW/(n-k)} \ge F_\alpha(k-1,n-k).$$

**Proof.** $\hat{\theta}^\Omega = (\bar{x}_1,\ldots,\bar{x}_k,\hat{\sigma}^{2,\Omega})$, $n\hat{\sigma}^{2,\Omega} = SSW$; $\hat{\theta}^{\Omega_0} = (\bar{x},\ldots,\bar{x},\hat{\sigma}^{2,0})$, $n\hat{\sigma}^{2,0} = SSW + SSB$. So

$$2\{l(\hat{\theta}^\Omega) - l(\hat{\theta}^{\Omega_0})\} = n\log\frac{\hat{\sigma}^{2,0}}{\hat{\sigma}^{2,\Omega}} = n\log\Big(1 + \frac{SSB}{SSW}\Big),$$

increasing in $F_n$. The map $A : \mathbf{X} \mapsto (\bar{X}_i - \bar{X})$ is symmetric idempotent with $SSB = \mathbf{X}'A\mathbf{X}$, $\mathrm{tr}A = \sum_{i,j}(1/n_i - 1/n) = k-1$, $A\mathbf{1} = \mathbf{0}$; under $H_0$, $SSB/\sigma^2 = \mathbf{Z}'A\mathbf{Z} \sim \chi^2(k-1)$, $\mathbf{Z} = (\mathbf{X} - \bar{\mu}\mathbf{1})/\sigma$, independent of $SSW/\sigma^2 \sim \chi^2(n-k)$ (Theorem 10.2). ■

With unweighted effects $\tilde{\alpha}_i = \mu_i - \sum\mu_i/k$, the null is the same and contrasts agree: $\sum c_i\tilde{\alpha}_i = \sum c_i\alpha_i$.

**Example 10.1**

Yields (in kg) of a crop under three fertilizers:

| fertilizer | observations | $n_i$ | $\bar{x}_i$ |
|---|---|---|---|
| A | 20, 22, 19, 23 | 4 | 21 |
| B | 25, 27, 26 | 3 | 26 |
| C | 22, 21, 24, 23, 25 | 5 | 23 |

$n = 12$, $\bar{x} = 23.083$, $SSB = 42.92$, $SSW = 22$: $F = (42.92/2)/(22/9) = 8.78 > F_{0.05}(2,9) = 4.26$. Bonferroni ($m = 3$, $t_{0.05/6}(9) = 2.933$, $\hat{\sigma}^2 = 2.444$) for $\mu_B - \mu_A$: $5 \pm 2.933\sqrt{2.444(1/3+1/4)} = 5 \pm 3.50$, excluding 0.

## 10.2 Two-way analysis of variance

Factors $A$ ($i \le a$) and $B$ ($j \le b$); cell $(i,j)$ observed $n_{ij}$ times:

$$X_{ijk} = \mu_{ij} + e_{ijk}, \qquad e_{ijk} \overset{\text{i.i.d.}}{\sim} N(0,\sigma^2), \qquad k = 1,\dots,n_{ij}.$$

A one-way model with $ab$ treatments, so:

**Theorem 10.6**

(a) The UMVUEs are $\hat{\mu}_{ij} = \bar{X}_{ij}$ and $\hat{\sigma}^2 = \sum_{i,j,k}(X_{ijk} - \bar{X}_{ij})^2/(n-ab)$, $n = \sum n_{ij}$.
(b) The $\bar{X}_{ij}$ are independent $N(\mu_{ij}, \sigma^2/n_{ij})$.
(c) $\hat{\sigma}^2$ is independent of the cell means.
(d) $(n-ab)\hat{\sigma}^2/\sigma^2 \sim \chi^2(n-ab)$.
(e) Confidence sets and Scheffé intervals are as in Theorem 10.3, with $D = \operatorname{diag}(1/n_{ij})$ and $n - ab$ degrees of freedom.

The effects are defined by

$$\bar{\mu}_{\cdot\cdot} = \sum_{i,j} \frac{n_{ij}}{n}\mu_{ij}, \quad \bar{\mu}_{i\cdot} = \sum_j \frac{n_{ij}}{n_{i\cdot}}\mu_{ij}, \quad \bar{\mu}_{\cdot j} = \sum_i \frac{n_{ij}}{n_{\cdot j}}\mu_{ij},$$

$$\alpha_i = \bar{\mu}_{i\cdot} - \bar{\mu}_{\cdot\cdot}, \qquad \beta_j = \bar{\mu}_{\cdot j} - \bar{\mu}_{\cdot\cdot}, \qquad \gamma_{ij} = \mu_{ij} - \bar{\mu}_{i\cdot} - \bar{\mu}_{\cdot j} + \bar{\mu}_{\cdot\cdot}.$$

$\alpha_i, \beta_j$ are the **main effects**, $\gamma_{ij}$ (the non-additive part) the **interaction**. In the **balanced** case $n_{ij} \equiv r$,

$$X_{ijk} = \bar{\mu}_{\cdot\cdot} + \alpha_i + \beta_j + \gamma_{ij} + e_{ijk}, \qquad \sum_i \alpha_i = \sum_j \beta_j = \sum_i \gamma_{ij} = \sum_j \gamma_{ij} = 0.$$

**Theorem 10.7 (Pairwise comparisons of main effects)**

In the balanced model, $\hat{\alpha}_i = \bar{X}_{i\cdot\cdot} - \bar{X}_{\cdots}$, $n_{i\cdot} = rb$, $n = rab$, and

(a) with probability at least $1-\alpha$, for all $i \ne l$, $|(\alpha_i - \alpha_l) - (\hat{\alpha}_i - \hat{\alpha}_l)| \le \sqrt{2\hat{\sigma}^2/(rb)}\sqrt{(a-1)F_\alpha(a-1, n-ab)}$;
(b) with $m = a(a-1)/2$, $\alpha^* = \alpha/m$, the same holds with $t_{\alpha^*/2}(n-ab)$ in place of $\sqrt{(a-1)F_\alpha(a-1,n-ab)}$.

**Proof.** $\alpha_i - \alpha_l = \sum_j(\mu_{ij} - \mu_{lj})/b$ is a contrast with $\sum c^2/n_{ij} = 2/(rb)$, in an $(a-1)$-dimensional space; apply Theorem 10.6(e) and Theorem 10.4. ∎

**Theorem 10.8 (Test for interaction)**

In the balanced model let

$$SS_{AB} = \sum_{i,j} r(\bar{X}_{ij\cdot} - \bar{X}_{i\cdot\cdot} - \bar{X}_{\cdot j\cdot} + \bar{X}_{\cdots})^2, \qquad SSE = \sum_{i,j,k} (X_{ijk} - \bar{X}_{ij\cdot})^2.$$

The LRT of $H_0^{AB}: \gamma_{ij} = 0$ for all $i, j$ (the additive model) rejects when

$$F_n = \frac{SS_{AB}/\{(a-1)(b-1)\}}{SSE/(n-ab)} \geq F_\alpha((a-1)(b-1), n-ab).$$

**Proof.** Under $\Omega$, $\hat{\mu}_{ij} = \bar{x}_{ij\cdot}$, $n\hat{\sigma}^{2,\Omega} = SSE$; under $H_0^{AB}$ (balanced), $\hat{\mu}_{ij}^0 = \bar{x}_{i\cdot\cdot} + \bar{x}_{\cdot j\cdot} - \bar{x}_{\cdots}$ and by orthogonality $n\hat{\sigma}^{2,0} = SSE + SS_{AB}$, so the statistic is $n\log(1 + SS_{AB}/SSE)$. The map $\Pi: (x_{ijk}) \mapsto (\bar{x}_{ij\cdot} - \bar{x}_{i\cdot\cdot} - \bar{x}_{\cdot j\cdot} + \bar{x}_{\cdots})$ is symmetric idempotent, $SS_{AB} = \mathbf{X}'\Pi\mathbf{X}$, and

$$\mathrm{tr}\Pi = \sum_{i,j,k} \left(\frac{1}{r} - \frac{1}{rb} - \frac{1}{ra} + \frac{1}{rab}\right) = ab - a - b + 1 = (a-1)(b-1).$$

Under $H_0^{AB}$, $\Pi$ annihilates the mean, so $SS_{AB}/\sigma^2 \sim \chi^2((a-1)(b-1))$, a function of cell means, independent of $SSE/\sigma^2 \sim \chi^2(n-ab)$. ■

**Theorem 10.9 (Tests for main effects)**

In the balanced model, with $SS_A = \sum_i br(\bar{X}_{i\cdot\cdot} - \bar{X}_{\cdots})^2$ and $SS_B = \sum_j ar(\bar{X}_{\cdot j\cdot} - \bar{X}_{\cdots})^2$:

(a) the LRT of $H_0^A: \alpha_1 = \cdots = \alpha_a = 0$ rejects when $\dfrac{SS_A/(a-1)}{SSE/(n-ab)} \geq F_\alpha(a-1, n-ab)$;

(b) the LRT of $H_0^B: \beta_1 = \cdots = \beta_b = 0$ rejects when $\dfrac{SS_B/(b-1)}{SSE/(n-ab)} \geq F_\alpha(b-1, n-ab)$.

Proofs as for Theorem 10.8. Summary ($\delta$: noncentrality, next section):

| source | SS | df | MS | $F$ | $\delta$ |
|---|---|---|---|---|---|
| $A$ | $SS_A$ | $a-1$ | $MS_A$ | $MS_A/MSE$ | $\sum br\alpha_i^2/\sigma^2$ |
| $B$ | $SS_B$ | $b-1$ | $MS_B$ | $MS_B/MSE$ | $\sum ar\beta_j^2/\sigma^2$ |
| $A \times B$ | $SS_{AB}$ | $(a-1)(b-1)$ | $MS_{AB}$ | $MS_{AB}/MSE$ | $\sum r\gamma_{ij}^2/\sigma^2$ |
| error | $SSE$ | $n-ab$ | $MSE$ | | |
| total | $SST$ | $n-1$ | | | |

## 10.3 Power of $F$ tests

**Motivation.** Under alternatives $F$ is noncentral, and the noncentrality determines power and sample size.

**Theorem 10.10 (Noncentral chi-square)**

Let $X_i \sim N(\mu_i, 1)$, $i = 1, \ldots, r$, be independent, $Y = X_1^2 + \cdots + X_r^2$ and $\delta = \mu_1^2 + \cdots + \mu_r^2$. Then

(a) $K_Y(t) = -\frac{r}{2}\log(1-2t) + \frac{\delta}{2} \cdot \frac{2t}{1-2t} = \sum_{k\geq 1} \frac{t^k}{k!} 2^{k-1}(k-1)!\,(r + k\delta)$ for $t < 1/2$;

(b) with $p_{P(\delta/2)}(k)$ the Poisson$(\delta/2)$ pmf, $f_Y(y) = \sum_{k\ge 0} p_{P(\delta/2)}(k)\, f_{\chi^2(r+2k)}(y)$.

So the law depends only on $\delta$: the **noncentral chi-square** $\chi^2(r;\delta)$.

**Proof.** (a) For $X \sim N(\mu,1)$, $t < 1/2$, completing the square,

$$Ee^{tX^2} = \int \frac{1}{\sqrt{2\pi}} \exp\Big\{-\frac{1}{2}(1-2t)\Big(x - \frac{\mu}{1-2t}\Big)^2 + \frac{t\mu^2}{1-2t}\Big\}dx = (1-2t)^{-1/2}\exp\Big(\frac{\mu^2 t}{1-2t}\Big).$$

Multiply over $i$, take logs, and expand $-\log(1-2t) = \sum (2t)^k/k$, $\frac{2t}{1-2t} = \sum_{k\ge 1}(2t)^k$.

(b) By (a), $Y \overset{d}{=} V + Z^2$, $V \sim \chi^2(r-1)$ independent of $Z \sim N(\sqrt{\delta},1)$. With $\theta = \sqrt{\delta}$,

$$f_{Z^2}(x) = \frac{1}{2\sqrt{x}}\frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}x - \frac{1}{2}\theta^2}\big(e^{\theta\sqrt{x}} + e^{-\theta\sqrt{x}}\big) = \sum_{k\ge 0} p_{P(\delta/2)}(k) f_{\chi^2(1+2k)}(x),$$

expanding $e^{\theta\sqrt{x}} + e^{-\theta\sqrt{x}} = 2\sum_k \theta^{2k}x^k/(2k)!$ and using $\Gamma(\tfrac{1}{2})\Gamma(2k+1) = 2^{2k}\Gamma(k+\tfrac{1}{2})\Gamma(k+1)$; convolve with $\chi^2(r-1)$. ■

**Theorem 10.11**

If $\mathbf{X} \sim N(\mu, I)$ and $A$ is symmetric idempotent, then $\mathbf{X}'A\mathbf{X} \sim \chi^2(r;\delta)$ with $r = \operatorname{tr} A$ and $\delta = \mu' A \mu$.

**Proof.** $A = P\begin{pmatrix} I_r & 0 \\ 0 & 0 \end{pmatrix} P'$, $P$ orthogonal; $\mathbf{Z} = P'\mathbf{X} \sim N(\eta, I)$, $\eta = P'\mu$, and $\mathbf{X}'A\mathbf{X} = Z_1^2 + \cdots + Z_r^2$ with $\eta_1^2 + \cdots + \eta_r^2 = \mu' A \mu$. ■

For independent $V_1 \sim \chi^2(r_1;\delta)$, $V_2 \sim \chi^2(r_2)$, $(V_1/r_1)/(V_2/r_2) \sim$ **noncentral** $F$ $F(r_1, r_2; \delta)$.

**Example 10.2 (Power of the one-way $F$ test)**

$SSB = \mathbf{X}'A\mathbf{X}$ with $A$ idempotent of trace $k-1$, mapping $(\mu_i)$ to $(\alpha_i)$; by Theorem 10.11

$$SSB/\sigma^2 \sim \chi^2(k-1;\delta), \qquad \delta = \sum_i n_i \alpha_i^2/\sigma^2,$$

so the power is $P\{F(k-1, n-k;\delta) \ge F_\alpha(k-1, n-k)\}$. It increases in $\delta$: $F_n \overset{d}{=} \frac{(V+Z^2)/(k-1)}{W/(n-k)}$ ($V \sim \chi^2(k-2)$, $Z \sim N(\sqrt{\delta},1)$, $W \sim \chi^2(n-k)$ independent), so the power is $P_\theta(Z^2 \ge Y)$, $Y \perp Z$, $\theta = \sqrt{\delta}$, and

$$\frac{d}{d\theta} P_\theta(Z^2 \ge y) = e^{-y/2}\phi(\theta)\big(e^{\sqrt{y}\theta} - e^{-\sqrt{y}\theta}\big) > 0 \qquad (\theta > 0).$$

Likewise the balanced interaction test has power $P\{F((a-1)(b-1), n-ab; \delta_{AB}) \ge F_\alpha\}$, $\delta_{AB} = \sum r\gamma_{ij}^2/\sigma^2$.

## 10.4 Regression analysis

**Motivation.** Regression is the same linear-model theory with continuous regressors.

The normal linear regression model (Chapters 4, 6):

$$\mathbf{Y} = X\beta + \mathbf{e}, \qquad \mathbf{e} \sim N_n(\mathbf{0}, \sigma^2 I), \qquad \beta \in \mathbb{R}^{p+1},\ \sigma^2 > 0,\ \operatorname{rank} X = p+1.$$

**Theorem 10.12**

(a) The UMVUEs of $\mathbf{c}'\beta$ and $\sigma^2$ are $\mathbf{c}'\hat\beta = \mathbf{c}'(X'X)^{-1}X'\mathbf{Y}$ and $\hat\sigma^2 = \|\mathbf{Y}-X\hat\beta\|^2/(n-p-1)$.

(b) $\hat\beta \sim N_{p+1}(\beta, \sigma^2(X'X)^{-1})$.

(c) $\hat\beta$ and $\hat\sigma^2$ are independent.

(d) $(n-p-1)\hat\sigma^2/\sigma^2 \sim \chi^2(n-p-1)$.

**Proof.** The density is a full-rank exponential family, so $(X'\mathbf{Y}, \mathbf{Y}'\mathbf{Y})$ is complete sufficient; $X'\mathbf{Y} = (X'X)\hat\beta$ and $\mathbf{Y}'\mathbf{Y} = (n-p-1)\hat\sigma^2 + \hat\beta'(X'X)\hat\beta$, so the unbiased $\mathbf{c}'\hat\beta$, $\hat\sigma^2$ (Theorem 6.16) are UMVUEs. (b)–(d): Theorem 4.19 in Volume I. ■

**Theorem 10.13 (Confidence sets for $\beta$)**

For a $(p+1)\times r$ matrix $C$ of rank $r$:

(a) $P\{(C'\beta - C'\hat\beta)'(C'(X'X)^{-1}C)^{-1}(C'\beta - C'\hat\beta) \le r\hat\sigma^2 F_\alpha(r, n-p-1)\} = 1-\alpha$;

(b) $P\Big\{|\mathbf{c}'\beta - \mathbf{c}'\hat\beta| \le \sqrt{\mathbf{c}'(X'X)^{-1}\mathbf{c}\,\hat\sigma^2}\sqrt{rF_\alpha(r, n-p-1)} \;\; \forall \mathbf{c} \in \mathrm{col}(C)\Big\} = 1-\alpha$.

Proof as for Theorem 10.3.

To test a subset of coefficients (e.g. all slopes), partition

$$\mathbf{Y} = X_0\beta_0 + X_1\beta_1 + \mathbf{e}, \qquad \operatorname{rank} X_0 = p_0,\ \operatorname{rank}(X_0, X_1) = p_0 + p_1,$$

with $\Pi_0 = X_0(X_0'X_0)^{-1}X_0'$, $X_{1|0} = (I-\Pi_0)X_1$, $\Pi_{1|0} = X_{1|0}(X_{1|0}'X_{1|0})^{-1}X_{1|0}'$, $\Pi_{0,1} = X(X'X)^{-1}X' = \Pi_0 + \Pi_{1|0}$, $\Pi_0\Pi_{1|0} = 0$ (Theorem 6.17).

**Theorem 10.14 (Testing a subset of coefficients)**

For $H_0 : \beta_1 = \mathbf{0}$ vs $H_1 : \beta_1 \ne \mathbf{0}$:

(a) $\min_{\beta_0}\|\mathbf{Y} - X_0\beta_0\|^2 = \min_{\beta_0,\beta_1}\|\mathbf{Y} - X_0\beta_0 - X_1\beta_1\|^2 + \mathbf{Y}'\Pi_{1|0}\mathbf{Y}$.

(b) With $R(1|0) = \mathbf{Y}'\Pi_{1|0}\mathbf{Y}$ and $SSE = \mathbf{Y}'(I - \Pi_0 - \Pi_{1|0})\mathbf{Y}$, the LRT rejects when

$$F_n = \frac{R(1|0)/p_1}{SSE/(n-p_0-p_1)} \ge F_\alpha(p_1, n-p_0-p_1).$$

(c) $F_n \sim F(p_1, n-p_0-p_1; \delta)$ with $\delta = \beta_1'X_1'\Pi_{1|0}X_1\beta_1/\sigma^2$, and the power increases in $\delta$.

**Proof.** (a) The minima are $\mathbf{Y}'(I-\Pi_0)\mathbf{Y}$ and $\mathbf{Y}'(I-\Pi_{0,1})\mathbf{Y}$, with $I - \Pi_0 = (I-\Pi_{0,1}) + \Pi_{1|0}$.

(b) The MLEs of $\sigma^2$ are $SSE/n$ and $(SSE + R(1|0))/n$, giving $n\log(1 + R(1|0)/SSE)$.

(c) $\Pi_{1|0}$ is idempotent of trace $p_1$, so $R(1|0)/\sigma^2 \sim \chi^2(p_1;\delta)$, $\delta = \beta_1'X_1'\Pi_{1|0}X_1\beta_1/\sigma^2$ ($\Pi_{1|0}X_0 = 0$); $SSE/\sigma^2 \sim \chi^2(n-p_0-p_1)$, independent since $\Pi_{1|0}(I-\Pi_0-\Pi_{1|0}) = 0$. Monotone power as in Example 10.2. ■

**Theorem 10.15 (General linear hypotheses)**

For a $(p+1)\times r$ matrix $C$ of rank $r$, the LRT of $H_0 : C'\beta = \mathbf{0}$ vs $H_1 : C'\beta \ne \mathbf{0}$ satisfies:

(a) with $\Pi_{1|0} = X(X'X)^{-1}C\{C'(X'X)^{-1}C\}^{-1}C'(X'X)^{-1}X'$,

$$\min_{C'\beta=\mathbf{0}}\|\mathbf{Y} - X\beta\|^2 = \min_{\beta}\|\mathbf{Y} - X\beta\|^2 + \mathbf{Y}'\Pi_{1|0}\mathbf{Y},$$

$$\mathbf{Y}'\Pi_{1|0}\mathbf{Y} = (C'\hat\beta)'\{C'(X'X)^{-1}C\}^{-1}(C'\hat\beta);$$

(b) with $SSE = \mathbf{Y}'(I - X(X'X)^{-1}X')\mathbf{Y}$, it rejects when $\dfrac{\mathbf{Y}'\Pi_{1|0}\mathbf{Y}/r}{SSE/(n-p-1)} \geq F_\alpha(r, n-p-1)$;

(c) this statistic is $F(r, n-p-1; \delta)$ with $\delta = \beta' C\{C'(X'X)^{-1}C\}^{-1}C'\beta/\sigma^2$.

**Proof.** Reduce to Theorem 10.14: take orthonormal $C_1$ spanning $\mathrm{col}(C)$, extended to an orthonormal basis $(C_0\ C_1)$, so $C = C_1 B$, $B$ invertible. With $D_0 = XC_0$, $D_1 = XC_1$, $\gamma_0 = C_0'\beta$, $\gamma_1 = C_1'\beta$,

$$X\beta = X(C_0\ C_1)(C_0\ C_1)'\beta = D_0\gamma_0 + D_1\gamma_1,$$

and $C'\beta = \mathbf{0} \iff \gamma_1 = \mathbf{0}$. So $\Pi_{1|0}$ projects onto the complement of $\mathrm{col}(D_0)$ in $\mathrm{col}(X)$:

$$\{X\beta : \beta' X'XC_0 = \mathbf{0}\} = \{X\beta : X'X\beta \in \mathrm{col}(C)\} = \mathrm{col}(X(X'X)^{-1}C).$$

With $S = X(X'X)^{-1}C$, $\Pi_{1|0} = S(S'S)^{-1}S'$, and $S'\mathbf{Y} = C'\hat\beta$. ∎

**Linear models.** ANOVA models are regressions with dependent design columns, identified by constraints:

$$\mathbf{Y} = X\beta + \mathbf{e}, \quad C'\beta = \mathbf{0}, \quad \mathbf{e} \sim N_n(\mathbf{0}, \sigma^2 I), \quad \operatorname{rank} X = p + 1 - r < p + 1,$$

with the rows of $X$ and $C'$ jointly independent: a **linear model**.

**Example 10.3 (One-way model as a linear model)**

$X_{ij} = \bar\mu + \alpha_i + e_{ij}$, $\sum n_i\alpha_i = 0$: with $\mathbf{Y}$ stacked by groups, $\beta = (\bar\mu, \alpha_1, \ldots, \alpha_k)'$, $\mathbf{c} = (0, n_1, \ldots, n_k)'$,

$$X = \begin{pmatrix} \mathbf{1}_{n_1} & \mathbf{1}_{n_1} & \mathbf{0} & \cdots & \mathbf{0} \\ \mathbf{1}_{n_2} & \mathbf{0} & \mathbf{1}_{n_2} & \cdots & \mathbf{0} \\ \vdots & & & \ddots & \\ \mathbf{1}_{n_k} & \mathbf{0} & \mathbf{0} & \cdots & \mathbf{1}_{n_k} \end{pmatrix}.$$

The **design matrix** has rank $k < k+1$; $\mathbf{c}'\beta = 0$ identifies $\beta$. With $X'X$ singular, use projections instead of $(X'X)^{-1}$; the theory goes through as in Theorem 10.15.

# Chapter 11. Bayesian Inference

## 11.1 Prior and posterior distributions

Bayesian inference describes uncertainty about $\theta \in \Omega$ before the data by a **prior density** $\pi(\theta)$ (whose existence and uniqueness is the source of a long controversy; here it is given). We write $f(x \mid \theta)$ for the model density.

**Theorem 11.1 (Bayes' theorem)**

If $X_1, \dots, X_n$ is a sample from $f(x \mid \theta)$ and $\theta$ has prior density $\pi(\theta)$, then given $\mathbf{X} = \mathbf{x}$ the density of $\theta$ is

$$\pi(\theta \mid \mathbf{x}) \propto L(\theta; \mathbf{x})\pi(\theta), \qquad L(\theta; \mathbf{x}) = \prod_{i=1}^{n} f(x_i \mid \theta).$$

**Proof.** $\pi(\theta \mid \mathbf{x}) = f(\mathbf{x} \mid \theta)\pi(\theta)/f_{\mathbf{X}}(\mathbf{x})$ with $f_{\mathbf{X}}(\mathbf{x}) = \int_\Omega f(\mathbf{x} \mid \theta)\pi(\theta)\, d\theta$ free of $\theta$. ∎

$\pi(\theta \mid \mathbf{x})$ is the **posterior density**: posterior $\propto$ likelihood $\times$ prior. An **improper** prior ($\int \pi = \infty$) may be used if the posterior is proper.

**Example 11.1 (Binomial–beta)**

Bernoulli($\theta$) data with prior Beta($\alpha, \beta$): with $t = \sum x_i$,

$$\pi(\theta \mid \mathbf{x}) \propto \theta^t (1-\theta)^{n-t} \theta^{\alpha-1}(1-\theta)^{\beta-1},$$

so $\theta \mid \mathbf{x} \sim \text{Beta}(t+\alpha, n-t+\beta)$.

**Example 11.2 (Poisson–gamma)**

Poisson($\theta$) data with prior Gamma($\alpha, \beta$): $\pi(\theta \mid \mathbf{x}) \propto e^{-n\theta}\theta^t \theta^{\alpha-1} e^{-\theta/\beta}$, so $\theta \mid \mathbf{x} \sim \text{Gamma}(t+\alpha, (n+\beta^{-1})^{-1})$.

**Example 11.3 (Exponential–gamma)**

Exp($\theta$) data, rate $\lambda = 1/\theta \sim \text{Gamma}(\alpha, \beta)$: $\pi(\lambda \mid \mathbf{x}) \propto \lambda^n e^{-\lambda t}\lambda^{\alpha-1}e^{-\lambda/\beta}$, so $\lambda \mid \mathbf{x} \sim \text{Gamma}(n+\alpha, (t+\beta^{-1})^{-1})$; $\pi(\theta \mid \mathbf{x}) = \pi(\lambda \mid \mathbf{x})\theta^{-2}$.

**Example 11.4 (Normal mean, known variance)**

$N(\theta, \sigma^2)$ data, $\sigma^2$ known, prior $N(\mu_0, \sigma_0^2)$:

$$\pi(\theta \mid \mathbf{x}) \propto \exp\Big\{ -\frac{n}{2\sigma^2}(\theta - \bar{x})^2 - \frac{1}{2\sigma_0^2}(\theta - \mu_0)^2 \Big\}.$$

Completing the square, $\theta \mid \mathbf{x} \sim N(\mu(\mathbf{x}), \sigma^2(\mathbf{x}))$ with

$$\mu(\mathbf{x}) = \frac{n\sigma^{-2}\bar{x} + \sigma_0^{-2}\mu_0}{n\sigma^{-2} + \sigma_0^{-2}}, \qquad \sigma^2(\mathbf{x}) = (n\sigma^{-2} + \sigma_0^{-2})^{-1}.$$

Posterior precision = data precision + prior precision.

A prior whose posterior stays in its family is **conjugate**.

**Theorem 11.2 (Normal model, conjugate prior)**

Let $X_1, \dots, X_n$ be $N(\mu, \sigma^2)$, and let the prior of $\theta = (\mu, \sigma^2)$ be

$$\mu \mid \sigma^2 \sim N(\mu_0, \sigma^2/\kappa_0), \qquad \nu_0\sigma_0^2/\sigma^2 \sim \chi^2(\nu_0).$$

Put $\kappa_n = n + \kappa_0$, $\nu_n = n + \nu_0$, $s^2 = \sum(x_i - \bar{x})^2/(n-1)$, and

$$\mu_n(\mathbf{x}) = \frac{n\bar{x} + \kappa_0\mu_0}{n + \kappa_0}, \qquad \nu_n\sigma_n^2(\mathbf{x}) = \nu_0\sigma_0^2 + (n-1)s^2 + (\kappa_0^{-1} + n^{-1})^{-1}(\bar{x} - \mu_0)^2.$$

Then

(a) $\mu \mid \sigma^2, \mathbf{x} \sim N(\mu_n(\mathbf{x}), \sigma^2/\kappa_n)$ and $\nu_n\sigma_n^2(\mathbf{x})/\sigma^2 \mid \mathbf{x} \sim \chi^2(\nu_n)$;

(b) $\dfrac{\mu - \mu_n(\mathbf{x})}{\sigma_n(\mathbf{x})/\sqrt{\kappa_n}} \,\Big|\, \mathbf{x} \sim t(\nu_n)$.

**Proof.** (a) With $\tau = \sigma^{-2}$,

$$L \propto \tau^{n/2} \exp\Big\{ -\frac{\tau}{2}\big[(n-1)s^2 + n(\mu - \bar{x})^2\big]\Big\},$$

$$\pi(\mu \mid \tau) \propto \tau^{1/2} e^{-\frac{\tau\kappa_0}{2}(\mu-\mu_0)^2}, \qquad \pi(\tau) \propto \tau^{\nu_0/2-1} e^{-\frac{\tau}{2}\nu_0\sigma_0^2}.$$

Completing the square in $\mu$,

$$n(\mu - \bar{x})^2 + \kappa_0(\mu - \mu_0)^2 = \kappa_n(\mu - \mu_n(\mathbf{x}))^2 + \frac{n\kappa_0}{n + \kappa_0}(\bar{x} - \mu_0)^2,$$

with $\frac{n\kappa_0}{n+\kappa_0} = (\kappa_0^{-1} + n^{-1})^{-1}$, so

$$\pi(\mu, \tau \mid \mathbf{x}) \propto \tau^{1/2} e^{-\frac{\tau\kappa_n}{2}(\mu-\mu_n(\mathbf{x}))^2} \cdot \tau^{\nu_n/2-1} e^{-\frac{\tau}{2}\nu_n\sigma_n^2(\mathbf{x})},$$

(b) Given $\mathbf{x}$ and $\sigma^2$, $Z = \frac{\mu-\mu_n(\mathbf{x})}{\sigma/\sqrt{\kappa_n}} \sim N(0,1)$, free of $\sigma^2$, so $Z \perp V = \nu_n\sigma_n^2(\mathbf{x})/\sigma^2 \sim \chi^2(\nu_n)$, and $Z/\sqrt{V/\nu_n} \sim t(\nu_n)$. ∎

Without conjugacy $f_{\mathbf{X}}(\mathbf{x})$ is often intractable and MCMC is used; we keep conjugate priors.

## 11.2 Bayesian inference

Summarize the posterior: **posterior mean**, **median** or **mode**, with posterior variance or quantiles. $I(\mathbf{x})$ with $P(\theta \in I(\mathbf{x}) \mid \mathbf{X} = \mathbf{x}) = 1 - \alpha$ is a **credible interval**, a probability statement about $\theta$ given the data.

**Example 11.5 (Normal mean, known variance)**

The normal posterior has equal mean, median and mode:

$$\hat{\theta} = \frac{n\sigma^{-2}\bar{x} + \sigma_0^{-2}\mu_0}{n\sigma^{-2} + \sigma_0^{-2}},$$

weighting $\bar{x}$ and $\mu_0$ by data precision $n\sigma^{-2}$ and prior precision $\sigma_0^{-2}$; credible interval $\hat{\theta} \pm z_{\alpha/2}\sigma(\mathbf{x})$. E.g. $\sigma = 10$, $n = 25$, $\bar{x} = 104$, prior $N(100, 5^2)$: $\hat{\theta} = (26 + 4)/0.29 = 103.45$, $\sigma(\mathbf{x}) = 1.857$, 95% interval $103.45 \pm 3.64$.

**Example 11.6 (Normal model, conjugate prior)**

By Theorem 11.2(b), for $\nu_n > 1$ the posterior of $\mu$ is a shifted, scaled $t$: mean, median and mode $\hat{\mu} = \mu_n(\mathbf{x})$, interval $\hat{\mu} \pm t_{\alpha/2}(\nu_n)\sigma_n(\mathbf{x})/\sqrt{\kappa_n}$.
For $\sigma^2$, with $Y = \nu_n\sigma_n^2(\mathbf{x})/\sigma^2 \mid \mathbf{x} \sim \chi^2(\nu_n)$,

$$\hat{\sigma}^{2,\text{mean}} = \nu_n\sigma_n^2(\mathbf{x})E[Y^{-1}] = \frac{\nu_n}{\nu_n - 2}\sigma_n^2(\mathbf{x}), \qquad \hat{\sigma}^{2,\text{median}} = \frac{\nu_n}{\chi^2_{0.5}(\nu_n)}\sigma_n^2(\mathbf{x}),$$

and $\hat{\sigma}^{2,\text{mode}} = \frac{\nu_n}{\nu_n+2}\sigma_n^2(\mathbf{x})$; all factors $\to 1$. The credible interval is

$$\frac{\nu_n\sigma_n^2(\mathbf{x})}{\chi^2_{\alpha/2}(\nu_n)} \le \sigma^2 \le \frac{\nu_n\sigma_n^2(\mathbf{x})}{\chi^2_{1-\alpha/2}(\nu_n)}.$$

**Example 11.7 (Binomial–beta)**

Prior $\text{Beta}(\alpha_1, \alpha_2)$, posterior $\text{Beta}(t + \alpha_1, n - t + \alpha_2)$, mean

$$\hat{\theta}^{\text{mean}} = \frac{t + \alpha_1}{n + \alpha_1 + \alpha_2} = \frac{n}{n + \alpha_1 + \alpha_2} \cdot \frac{t}{n} + \frac{\alpha_1 + \alpha_2}{n + \alpha_1 + \alpha_2} \cdot \frac{\alpha_1}{\alpha_1 + \alpha_2},$$

a weighted average of $t/n$ and the prior mean; mode $\frac{(t+\alpha_1-1)_+}{(t+\alpha_1-1)_+ + (n-t+\alpha_2-1)_+}$. The credible interval $[B_{1-\alpha/2}, B_{\alpha/2}]$ uses upper Beta quantiles, from $F$ tables via $\text{Beta}(r_1, r_2) \overset{d}{=} \frac{(r_1/r_2)F}{1+(r_1/r_2)F}$, $F \sim F(2r_1, 2r_2)$; for large $n$, $\hat{\theta} \pm z_{\alpha/2}\sqrt{\hat{\theta}(1-\hat{\theta})/(n + \alpha_1 + \alpha_2)}$. E.g. $t = 12$, $n = 40$, uniform prior: $\hat{\theta}^{\text{mean}} = 13/42 = 0.310$, interval $0.310 \pm 0.140$.

**Bayesian testing.** Compare the **posterior odds**

$$\frac{P(\theta \in \Omega_0 \mid \mathbf{X} = \mathbf{x})}{P(\theta \in \Omega_1 \mid \mathbf{X} = \mathbf{x})} = \frac{P(\theta \in \Omega_0 \mid \mathbf{X} = \mathbf{x})}{1 - P(\theta \in \Omega_0 \mid \mathbf{X} = \mathbf{x})}.$$

**Example 11.8 (Simple hypotheses)**

For $H_0 : \theta = \theta_0$ vs $H_1 : \theta = \theta_1$ with prior probabilities $\pi_0, \pi_1$,

$$\frac{P(\theta = \theta_0 \mid \mathbf{x})}{P(\theta = \theta_1 \mid \mathbf{x})} = \frac{f(\mathbf{x} \mid \theta_0)\pi_0}{f(\mathbf{x} \mid \theta_1)\pi_1}.$$

Rejecting when the odds are below 1 is the Bayes test of Theorem 9.1; for $N(\mu, 1)$ the odds are $\exp\{n(\mu_0 - \mu_1)(\bar{x} - \frac{\mu_0+\mu_1}{2}) + \log(\pi_0/\pi_1)\}$.

**Example 11.9 (One-sided hypotheses)**

(a) Normal model, conjugate prior, $H_0 : \mu \le \mu_0$: by Theorem 11.2(b)

$$P(\mu \le \mu_0 \mid \mathbf{x}) = F_{t(\nu_n)}\Big(\frac{\mu_0 - \mu_n(\mathbf{x})}{\sigma_n(\mathbf{x})/\sqrt{\kappa_n}}\Big),$$

($F_{t(\nu)}$ the $t(\nu)$ cdf).
(b) Binomial–beta, $H_0 : \theta \le \theta_0$: $P(\theta \le \theta_0 \mid \mathbf{x})$ is the posterior Beta cdf at $\theta_0$, or approximately $\Phi((\theta_0 - \hat{\theta})/\sqrt{\hat{\theta}(1-\hat{\theta})/(n + \alpha_1 + \alpha_2)})$.

## 11.3 Bayes and minimax estimators

Frequentist inference evaluates procedures over repeated samples, e.g. by MSE; as in Chapter 8, the Bayes and maximum MSE are free of $\theta$:

$$r(\pi, \hat{\theta}) = \int_\Omega E[(\hat{\theta} - \theta)^2 \mid \theta]\pi(\theta)\, d\theta, \qquad \max_{\theta \in \Omega} E[(\hat{\theta} - \theta)^2 \mid \theta].$$

**Definition (Bayes and minimax estimators)**

(a) $\hat{\theta}^\pi$ is a **Bayes estimator** for the prior $\pi$ if $r(\pi, \hat{\theta}^\pi) = \min_{\hat{\theta}} r(\pi, \hat{\theta})$.
(b) $\hat{\theta}^m$ is a **minimax estimator** if $\max_\theta E[(\hat{\theta}^m - \theta)^2 \mid \theta] = \min_{\hat{\theta}} \max_\theta E[(\hat{\theta} - \theta)^2 \mid \theta]$.

**Theorem 11.3 (Bayes estimator under squared error)**

The Bayes estimator minimizes the posterior expected squared error and is the posterior mean:

$$\hat{\theta}^\pi(\mathbf{x}) = \arg\min_a E[(a - \theta)^2 \mid \mathbf{X} = \mathbf{x}] = E(\theta \mid \mathbf{X} = \mathbf{x}).$$

**Proof.** By iterated expectation

$$r(\pi, \hat{\theta}) = E[E\{(\hat{\theta}(\mathbf{X}) - \theta)^2 \mid \mathbf{X}\}].$$

so pointwise minimization of the inner conditional expectation minimizes $r$; and $E[(a - \theta)^2 \mid \mathbf{x}] = \mathrm{Var}(\theta \mid \mathbf{x}) + (a - E[\theta \mid \mathbf{x}])^2$. ■

Likewise for a **loss function** $L(\theta, \hat{\eta})$: the **Bayes risk** $r(\pi, \hat{\eta}) = E[L(\theta, \hat{\eta}(\mathbf{X}))]$ is minimized by minimizing the **posterior expected loss** for each $\mathbf{x}$; minimax uses the **risk function** $R_{\hat{\eta}}(\theta) = E[L(\theta, \hat{\eta}) \mid \theta]$.

**Example 11.10**

(a) Binomial–beta with squared error: $\hat{\theta}^\pi = \frac{t + \alpha_1}{n + \alpha_1 + \alpha_2}$.
(b) Normal model, conjugate prior, $L = (\hat{\mu} - \mu)^2/\sigma^2$: $\hat{\mu}^\pi = E(\mu\sigma^{-2} \mid \mathbf{x})/E(\sigma^{-2} \mid \mathbf{x}) = \mu_n(\mathbf{x})$, since $E(\mu \mid \sigma^2, \mathbf{x}) = \mu_n(\mathbf{x})$ is free of $\sigma^2$.
(c) $L = (\hat{\sigma}^2/\sigma^2 - 1)^2$: $\hat{\sigma}^{2,\pi} = E(\sigma^{-2} \mid \mathbf{x})/E(\sigma^{-4} \mid \mathbf{x})$, and with $EY = \nu_n$, $EY^2 = 2\nu_n + \nu_n^2$,

$$\hat{\sigma}^{2,\pi} = \nu_n \sigma_n^2(\mathbf{x}) \frac{\nu_n}{2\nu_n + \nu_n^2} = \frac{\nu_n}{\nu_n + 2} \sigma_n^2(\mathbf{x}),$$

the posterior mode.

**Theorem 11.4 (Bayes estimators with constant risk are minimax)**

If an estimator $\hat{\eta}^*$ has (a) constant risk, $R_{\hat{\eta}^*}(\theta) = R_{\hat{\eta}^*}(\theta_0)$ for all $\theta$, and (b) is the Bayes estimator for some prior probability distribution $\pi$, then it is minimax.

**Proof.** By (a), $\max_\theta R_{\hat{\eta}^*} = r(\pi, \hat{\eta}^*)$; by (b), for any $\hat{\eta}$,

$$r(\pi, \hat{\eta}^*) \le r(\pi, \hat{\eta}) = \int R_{\hat{\eta}}(\theta)\pi(\theta)\, d\theta \le \max_\theta R_{\hat{\eta}}(\theta).$$

■

**Example 11.11 (Minimax estimation of a binomial proportion)**

Bernoulli($\theta$), squared error, estimators $a\hat\theta + b$, $\hat\theta = t/n$:

$$E[(a\hat\theta + b - \theta)^2 \mid \theta] = \frac{a^2\theta(1-\theta)}{n} + \{b + (a-1)\theta\}^2,$$

the $\theta^2$ and $\theta$ coefficients vanish when $(1-a)^2 = a^2/n$ and $a^2/n - 2b(1-a) = 0$:

$$a = \frac{\sqrt{n}}{\sqrt{n}+1}, \qquad b = \frac{1}{2(\sqrt{n}+1)}.$$

$\hat\theta^* = \frac{t+\sqrt{n}/2}{n+\sqrt{n}}$ has constant risk $\frac{1}{4(\sqrt{n}+1)^2}$ and is Bayes for Beta($\sqrt{n}/2, \sqrt{n}/2$) (Example 11.10(a)): minimax (Theorem 11.4). Its maximum risk is below $1/(4n)$, but $\hat\theta$ is better except near $\theta = 1/2$.

A minimax estimator may only be a limit of Bayes estimators:

**Theorem 11.5 (Least favourable sequence of priors)**

If for some estimator $\hat\eta^*$ there are priors $\pi_k$ with Bayes estimators $\hat\eta^{\pi_k}$ such that

$$\lim_{k\to\infty} r(\pi_k, \hat\eta^{\pi_k}) \geq \max_\theta R_{\hat\eta^*}(\theta),$$

then $\hat\eta^*$ is minimax.

**Proof.** For any $\hat\eta$, $r(\pi_k, \hat\eta^{\pi_k}) \leq r(\pi_k, \hat\eta) \leq \max_\theta R_{\hat\eta}$; let $k \to \infty$. ■

**Example 11.12 (The sample mean is minimax)**

$N(\mu, \sigma^2)$, loss $(\hat\mu - \mu)^2/\sigma^2$: the conjugate Bayes estimator $\mu_n(\mathbf{x})$ has Bayes risk $1/\kappa_n = 1/(n + \kappa_0)$. With $\kappa_0 = 1/k$, $r(\pi_k, \hat\mu^{\pi_k}) \to 1/n$, the constant risk of $\bar X$; by Theorem 11.5 $\bar X$ is minimax.

## 11.4 Selected exercises

**Motivation.** Book exercises: two more conjugate families, a Bayes estimator under relative loss, and a weighted loss making $\bar X$ Bayes and minimax.

Gamma($\alpha, \beta$) has density $\theta^{\alpha-1}e^{-\theta/\beta}/\{\Gamma(\alpha)\beta^\alpha\}$ (Example 11.2).

**Exercise 11.1 (Pareto data, gamma prior)**

Let $X_1, \dots, X_n$ have density $f(x;\theta) = \theta x^{-\theta-1}$ for $x > 1$ (Pareto$(1,\theta)$, $\theta > 0$), and let $\theta \sim$ Gamma$(\alpha, \beta)$. Find the posterior distribution of $\theta$.

**Solution.**

*Step 1.* With $t = \sum_i \log x_i > 0$, $L(\theta; \mathbf{x}) \propto \theta^n e^{-\theta t}$.

*Step 2.* By Theorem 11.1,

$$\pi(\theta \mid \mathbf{x}) \propto \theta^n e^{-\theta t} \cdot \theta^{\alpha-1} e^{-\theta/\beta} = \theta^{n+\alpha-1} e^{-\theta(t+\beta^{-1})},$$

so $\theta \mid \mathbf{x} \sim$ Gamma$\big(n+\alpha, (t+\beta^{-1})^{-1}\big)$ (Example 11.3: $\log X_i$ is exponential with rate $\theta$). ■

**Exercise 11.2 (Multinomial data, Dirichlet prior)**

Let $\mathbf{X} = (X_1, \dots, X_k)' \sim \text{Multi}(n, (p_1, \dots, p_k)')$ with $\sum_j p_j = 1$, and let $\theta = (p_1, \dots, p_{k-1})'$ have the Dirichlet$(\alpha_1, \dots, \alpha_k)$ prior, whose density on $\{p_j > 0,\ \sum_{j<k} p_j < 1\}$ is

$$\pi(\theta) = \frac{\Gamma(\alpha_1 + \cdots + \alpha_k)}{\Gamma(\alpha_1) \cdots \Gamma(\alpha_k)} \prod_{j=1}^{k} p_j^{\alpha_j - 1}, \qquad p_k = 1 - \sum_{j<k} p_j.$$

Find the posterior distribution of $\theta$.

**Solution.**

*Step 1.* $L(\theta; \mathbf{x}) \propto \prod_j p_j^{x_j}$, $p_k = 1 - \sum_{j<k} p_j$.

*Step 2.* By Theorem 11.1, on the simplex,

$$\pi(\theta \mid \mathbf{x}) \propto \prod_{j=1}^{k} p_j^{x_j} \prod_{j=1}^{k} p_j^{\alpha_j - 1} = \prod_{j=1}^{k} p_j^{x_j + \alpha_j - 1},$$

the Dirichlet$(x_1 + \alpha_1, \dots, x_k + \alpha_k)$ kernel; so this is the posterior. For $k = 2$ it is Example 11.1; $\alpha_j$ act as prior counts. ■

**Exercise 11.10 (A Bayes estimator under relative squared error)**

In Exercise 11.1, find the Bayes estimator of $\theta$ for the loss $L(\theta, \hat{\theta}) = (\hat{\theta}/\theta - 1)^2$, assuming $n + \alpha > 2$.

**Solution.**

*Step 1.* Minimize the posterior expected loss (after Theorem 11.3):

$$E\left[(a/\theta - 1)^2 \mid \mathbf{x}\right] = a^2 E(\theta^{-2} \mid \mathbf{x}) - 2aE(\theta^{-1} \mid \mathbf{x}) + 1,$$

minimized at $a^* = E(\theta^{-1} \mid \mathbf{x}) / E(\theta^{-2} \mid \mathbf{x})$.

*Step 2.* For $\theta \sim \text{Gamma}(a_1, b)$, $k < a_1$,

$$E\theta^{-k} = \frac{\int_0^\infty \theta^{a_1 - k - 1} e^{-\theta/b}\, d\theta}{\Gamma(a_1) b^{a_1}} = \frac{\Gamma(a_1 - k)}{\Gamma(a_1)}\, b^{-k}.$$

so $E\theta^{-1} = \{(a_1 - 1)b\}^{-1}$, $E\theta^{-2} = \{(a_1 - 1)(a_1 - 2)b^2\}^{-1}$ $(a_1 > 2)$.

*Step 3.* With $a_1 = n + \alpha$, $b = (t + \beta^{-1})^{-1}$ (Exercise 11.1),

$$\hat{\theta}^{\pi} = (a_1 - 2)b = \frac{n + \alpha - 2}{\sum_i \log x_i + \beta^{-1}}.$$

smaller than the posterior mean $(n + \alpha)b$: relative loss penalizes overestimating small $\theta$. ■

**Exercise 11.12 (A weighted loss for a proportion)**

Let $X_1, \dots, X_n$ be Bernoulli$(\theta)$, $0 < \theta < 1$, with loss $L(\theta, \hat{\theta}) = (\hat{\theta} - \theta)^2 / \{\theta(1 - \theta)\}$.
(a) Find the Bayes estimator for the uniform prior $U(0, 1)$.
(b) Find a minimax estimator of $\theta$.

**Solution.**

(a)

*Step 1.* The posterior is $\text{Beta}(t+1, n-t+1)$ (Example 11.1), so the posterior expected loss of $a$ is $\propto$

$$Q(a) = \int_0^1 (a-\theta)^2 \theta^{t-1}(1-\theta)^{n-t-1}\, d\theta.$$

*Step 2.* For $0 < t < n$, $Q(a) = a^2 B(t, n-t) - 2aB(t+1, n-t) + B(t+2, n-t)$ is minimized at

$$a^* = \frac{B(t+1,n-t)}{B(t,n-t)} = \frac{\Gamma(t+1)\Gamma(n)}{\Gamma(t)\Gamma(n+1)} = \frac{t}{n}.$$

*Step 3.* For $t = 0$, the integrand is $\sim a^2\theta^{-1}$ near 0, so $Q(a) = \infty$ unless $a = 0$, and $Q(0) < \infty$: $a^* = 0 = t/n$; symmetrically $a^* = 1$ for $t = n$.

*Step 4.* So $\hat{\theta}^\pi = \bar{X}$.

(b) $\bar{X}$ has risk

$$E\Big[\frac{(\bar{X}-\theta)^2}{\theta(1-\theta)} \;\Big|\; \theta\Big] = \frac{\text{Var}(\bar{X} \mid \theta)}{\theta(1-\theta)} = \frac{\theta(1-\theta)/n}{\theta(1-\theta)} = \frac{1}{n},$$

constant; with (a) and Theorem 11.4, $\bar{X}$ is minimax. (Under unweighted squared error it is not, Example 11.11; the weight equalizes its risk.) ■